Geopolitical Risk and Monetary Policy Transmission: Evidence from Egypt, Turkey, and Israel

By

Omar Ben Kaddour

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JOSEPH WHARTON SCHOLARS

Faculty Advisor:

Itay Goldstein

Joel S. Ehrenkranz Family Professor, Finance Department

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I. Introduction

Geopolitical risk (GPR) has become a first-order macro-financial shock. Episodes of war, terrorism, interstate tensions, and domestic political instability can trigger abrupt repricing in financial markets, shifts in capital flows, and changes in households' and firms' willingness to spend and invest. A growing literature formalizes by showing that adverse GPR shocks are associated, on average, with weaker real activity and tighter financial conditions, consistent with transmission through uncertainty, time-varying risk premia, and precautionary behavior (Caldara and Iacoviello, 2022a). At the same time, central banks remain the primary stabilization institutions tasked with preserving inflation control and macroeconomic credibility when such shocks occur.

This thesis studies a simple but policy-relevant question: when geopolitical risk is high, does monetary policy transmit to the economy in the same way as it does during normal times? In standard macroeconomic frameworks, changes in the policy rate affect borrowing costs, asset prices, credit creation, and exchange rates, which then shape inflation and real activity. Under geopolitical stress, however, these linkages may weaken or re-route. Investors may demand larger compensation for holding domestic assets; banks may tighten lending standards independently of the policy stance; and exchange-rate dynamics may be driven less by interest differentials and more by flight-to-safety behavior and external financing conditions. If these mechanisms are operative, a policy-rate increase of a given size can have different effects on sovereign risk premia, the currency, domestic credit, and ultimately output and inflation depending on the geopolitical environment.

Existing research motivates this interaction but does not fully resolve it. Work on geopolitical risk establishes GPR as an uncertainty-like disturbance with measurable macroeconomic and financial effects (Caldara and Iacoviello, 2022a), and related measures of political risk at the firm level show economically meaningful real and financial responses (Hassan et al., 2019). Separately, a large monetary-policy transmission literature—using narrative shocks, futures-based surprises, and high-frequency identification—characterizes average effects of policy on financial conditions and real outcomes (Romer and Romer, 2004; Gertler and Karadi, 2015). In open-economy settings, external financial conditions and global risk appetite can dominate domestic interest differentials, constraining monetary autonomy (Rey, 2015; Miranda-Agrippino and Rey, 2020; Bruno and Shin, 2015). There is also related work examining how central banks respond to geopolitical-risk shocks (Salachas et al., 2025; Ginn and Saadaoui, 2025). Yet comparatively little evidence directly quantifies how geopolitical stress reshapes the *strength* and *timing* of monetary-policy transmission, particularly in emerging markets where sovereign risk and external financing conditions are central to the policy problem.

The Middle East and North Africa, and the broader Eastern Mediterranean region, provide a natural setting in which to study this question. Countries such as Egypt, Turkey, and Israel have experienced recurrent geopolitical events over the past two decades, with frequent episodes of elevated security risk and regional tension. These economies also differ in important structural and institutional ways—exchange-rate regimes, inflation histories, financial openness, and policy credibility—creating useful cross-country variation in how shocks propagate. For Egypt and Turkey, risk-driven movements in sovereign spreads and the exchange rate can rapidly feed into domestic prices and credit conditions. For Israel, a

more institutionally mature monetary framework and deeper markets may imply a different interaction between geopolitical risk, asset pricing, and policy transmission. Examining these countries side-by-side therefore offers a high-stakes policy context and meaningful comparative leverage.

The contribution of this thesis is empirical and comparative. It measures monetary-policy innovations and traces their dynamic effects on key macro-financial outcomes—sovereign risk premia, interest rates along the yield curve, exchange rates, credit aggregates, inflation, and real activity—while allowing these responses to differ across geopolitical conditions. The goal is not only to estimate average impulse responses, but to characterize how transmission changes when risk is elevated, and whether the pattern is consistent with specific mechanisms: risk-premium amplification, impaired credit intermediation, or exchange-rate dominance under external stress.

The remainder of the thesis proceeds as follows. The next section reviews prior work on geopolitical risk and on monetary-policy transmission and develops testable hypotheses on how geopolitical stress should affect the asset-price, credit, and exchange-rate channels. Subsequent sections describe the empirical strategy and data construction, present pooled and country-level results, and then synthesize the evidence across countries and channels before concluding with limitations and directions for further research.

II. Literature Review

a) Geopolitical Risk Measurement and Economic Effects

Geopolitical risk refers to adverse events—including wars, terrorist attacks, military buildups, and diplomatic crises—that raise perceived threats to peace and security. A central contribution of the literature is to measure these tensions systematically. Caldara and Iacoviello (2022b) develop a newspaper-based **Geopolitical Risk (GPR)** index by counting coverage of geopolitical *threats* (e.g., war threats, nuclear tensions) and *acts* (e.g., conflict outbreaks, terrorist incidents) in leading international newspapers, scaled by overall news volume. The resulting monthly series spikes around major episodes of security stress (e.g., the World Wars, the Cuban Missile Crisis, and 9/11), providing an externally anchored proxy for global geopolitical risk as perceived by the public and markets (Caldara and Iacoviello, 2022b). By construction, GPR is conceptually distinct from broader uncertainty and volatility measures. For example, the **Economic Policy Uncertainty (EPU)** index of Baker et al. (2016) captures uncertainty about economic policy through policy-related news mentions, while the VIX reflects implied equity volatility and risk compensation in option prices. In contrast, GPR isolates geopolitical turmoil and displays only modest correlation with EPU and financial-volatility indices, indicating that it contains information not captured by standard uncertainty measures.(Caldara and Iacoviello, 2022b).

A consistent empirical finding is that heightened geopolitical risk is contractionary. Increases in GPR are associated with weaker investment, hiring, and confidence as firms and households postpone activity amid elevated uncertainty (Caldara and Iacoviello, 2022b). Caldara and Iacoviello (2022b) show that GPR increases predict declines in business fixed investment and employment over subsequent quarters, with effects arising both from the

threat of adverse events and from their *realization*. For emerging markets, Cheng and Chiu (2018) estimate sizable output contractions following global GPR shocks, highlighting the sensitivity of these economies to external risk conditions.

Geopolitical risk also has clear asset-market signatures that are directly relevant for policy transmission. Periods of elevated GPR are typically associated with "risk-off" behavior: equity drawdowns, retrenchment of capital flows, and higher required compensation for holding risky or illiquid claims. In fragile emerging equity markets, Hoque and Zaidi (2020) document large stock-market declines around major conflict events. In sovereign debt markets, Gamboa-Estrada and Romero (2025) show that emerging-market sovereign spreads and CDS-based risk premia widen significantly when global GPR rises. Smales (2021) further documents volatility spillovers across oil and equity markets around geopolitical shocks, consistent with broad portfolio rebalancing toward safe assets. Exchange rates provide an additional channel: by raising risk aversion and perceived country risk, geopolitical crises tend to put downward pressure on local currencies, and Caldara and Iacoviello (2022a) find systematic depreciation of emerging-market currencies in high-GPR months.

These patterns motivate why monetary-policy transmission may be state-dependent in periods of geopolitical stress. First, geopolitical risk can depress expected returns to real activity by raising uncertainty and weakening demand, while also discouraging cross-border investment and complicating trade relationships (Wang et al., 2024; Gupta et al., 2019). Second, geopolitical shocks can be inflationary through disruptions to energy and commodity supply, sanctions, and supply-chain frictions, generating cost pressures even as demand weakens (Yang et al., 2023; Caldara et al., 2025; Roscoe et al., 2022). Third, heightened GPR can directly alter financial conditions through portfolio rebalancing and capital-flow dynamics (Feng et al., 2023), including reduced foreign participation and weaker foreign direct investment (Wang et al., 2024). Fourth, geopolitical events may change trading conditions via tariffs and quotas, influencing exports and imports and thereby the external balance (Wang et al., 2020; Gupta et al., 2019). Finally, geopolitical risk can amplify financial instability and cross-market spillovers (Salachas et al., 2024; Li et al., 2022).

Taken together, these channels suggest that the mapping from a policy-rate change to sovereign spreads, the exchange rate, domestic credit, and ultimately output and inflation may differ in high- versus low-GPR environments. Under elevated risk, changes in real variables may be driven less by domestic interest differentials and forward guidance may play a larger role in shaping expectations and stabilizing confidence (Hansen et al., 2019).

b) Monetary Policy Transmission and State-Dependent Effects

In tranquil times, monetary policy influences the real economy through well-understood channels. First, the asset price channel (or interest rate channel) works via bond and equity markets: a policy rate cut, for example, lowers yields on government and corporate bonds, raising their prices; it also tends to boost stock prices by reducing the discount rate and improving profit outlooks. Higher asset valuations support spending through wealth effects and easier financing conditions. Second, the credit channel amplifies monetary policy via banks' lending behavior and borrowers' balance sheets Mishkin (1996). A lower policy rate can increase banks' willingness to lend (the bank lending channel) and improve borrowers' balance sheets by raising asset values and collateral (the balance-sheet channel), thereby

expanding credit to firms and households. Third, the exchange rate channel is often pivotal, especially in open economies: an interest rate hike attracts foreign capital and tends to appreciate the domestic currency (in a floating regime), which in turn affects net exports and inflation. Conversely, a rate cut usually leads to currency depreciation, boosting export competitiveness but also import prices. These transmission channels are well documented in advanced economies and many emerging markets Boivin and Giannoni (2006); Hofmann et al. (2016).

For instance, Hofmann et al. (2016) highlight a "risk-taking" aspect of the exchange-rate channel in emerging markets: when domestic currency appreciates (often as a result of tighter policy or positive risk sentiment), the local-currency yield curve can flatten and sovereign borrowing costs decline as foreign investors gain confidence in currency stability. This exemplifies how currency movements and sovereign yields interact in the transmission mechanism. In normal conditions, then, a monetary tightening should dampen credit growth, ease capital outflows (strengthen the FX rate), and temper inflation, while a loosening should do the opposite.

However, recent research suggests that the strength of these channels can vary dramatically depending on the state of the economy, including uncertainty and risk conditions. State-dependent transmission means that the effects of a given monetary policy shock may be amplified or dampened in certain environments. One pertinent factor is general economic uncertainty: Aastveit et al. (2017) show that when uncertainty is high, monetary policy shocks have significantly muted effects on output and investment. Intuitively, in a highly uncertain environment (for example, during a crisis or war scare), firms and consumers may be less responsive to interest rate changes – they might be more focused on risk management and liquidity hoarding than on borrowing or investing, blunting the usual channels. Aastveit et al. (2017) find that U.S. monetary easing in periods of elevated uncertainty yields a much smaller boost to aggregate demand than during stable periods, with the dampening most pronounced for investment spending. This aligns with theoretical expectations that precautionary behavior under uncertainty can weaken the traction of policy levers.

Cross-country evidence, likewise, indicates heterogeneity in monetary transmission related to structural differences. Deb et al. (2023) conduct a large cross-country analysis (33 advanced and emerging economies) and find substantial variation in the output and price responses to an identical monetary policy shock. Emerging market economies (EMEs) often exhibit faster and stronger exchange-rate responses – for instance, a rate hike in an EME may induce an immediate large appreciation – reflecting higher sensitivity of capital flows. But the pass-through from interest rates to domestic credit and demand can be weaker in some EMEs due to shallower financial systems or credibility issues.

Structural features such as the exchange rate regime play a key role: under a fixed or managed exchange rate, the conventional exchange-rate channel is suppressed, so more of the adjustment falls on reserves or other markets. In contrast, in economies with flexible exchange rates, currency movements can amplify the impact of policy on inflation (via import prices) and net exports. Banking sector depth and health also matter: in countries with well-capitalized banks and deep bond markets, a policy rate change more readily affects lending rates and asset prices, whereas in countries with financial frictions or extensive state-owned banks, the credit channel may be sluggish. These heterogeneities underscore that one size does not fit all – the "monetary transmission mechanism" differs across institutional

and macro-financial contexts Mishra et al. (2016).

In summary, the literature has developed a substantial understanding of (i) the adverse macroeconomic and financial effects of geopolitical risk and (ii) the mechanisms of monetary policy transmission, including their potential state dependence. What remains relatively underexplored is the intersection of these two strands: whether and how exogenous geopolitical risk shocks alter the propagation of monetary-policy shocks. By examining this interaction in the context of Egypt, Turkey, and Israel—three MENA economies frequently exposed to geopolitical stress—this thesis seeks to provide new empirical evidence on how geopolitical conditions condition monetary transmission.

III. Hypotheses

Let $MP_Shock_{c,t}$ denote an identified monetary policy shock for country c at month t, where $MP_Shock_{c,t} > 0$ is an unanticipated monetary tightening. Let $GPR_Shock_{c,t}$ denote an identified geopolitical risk shock. Outcomes include sovereign risk premia (e.g., CDS spreads), long yields $y10_{c,t}$, retail borrowing rates $LR_{c,t}$, the log exchange rate $s_{c,t}$ defined as local currency per USD (so $\Delta s_{c,t+1} > 0$ is a depreciation), credit aggregates, inflation, and real activity.

The baseline empirical design estimates state-dependent impulse responses via an interaction between $MP_Shock_{c,t}$ and $GPR_Shock_{c,t}$. Accordingly, the hypotheses are stated in terms of (i) the sign of the average response to a tightening shock and (ii) whether the magnitude and timing of that response change when $GPR_Shock_{c,t}$ is high.

H1 (Attenuated exchange-rate channel under high GPR). When $GPR_Shock_{c,t}$ is high, the exchange-rate response to monetary tightening is attenuated: the appreciation is smaller in magnitude (i.e., $\Delta s_{c,t+h}$ is less negative), and in extreme episodes the response may be near zero or positive (depreciation despite tightening).

H2 (Impaired pass-through to private borrowing costs under high GPR). When $GPR_Shock_{c,t}$ is high, the pass-through from monetary tightening to effective borrowing rates is weaker: lending rates $LR_{c,t}$ respond less per unit of $MP_Shock_{c,t}$ because the external finance premium and/or bank risk constraints dominate marginal rate-setting.

H3 (Greater real-side contraction under high GPR). When $GPR_Shock_{c,t}$ is high, a monetary tightening produces a larger decline in real activity than in low-risk periods. The mechanism is that geopolitical stress tightens financial conditions through risk premia and precautionary behavior, so a tightening shock delivered in a high-risk environment compounds the contractionary impulse.

H4 (Risk-premium amplification). When $GPR_Shock_{c,t}$ is high, sovereign risk premia (e.g., CDS spreads) increase more following a monetary tightening than in low-risk periods, reflecting the interaction of higher perceived default risk, weaker growth expectations, and reduced risk-bearing capacity.

IV. Data and Variable Construction

IV.A Sample, Frequency, and Country Coverage

We focus on three countries – **Egypt**, **Turkey**, and **Israel** – over the period 2011 through 2024, at a monthly frequency. These countries provide an ideal laboratory given their exposure to significant geopolitical events in the last decade (Arab Spring unrest, Syrian civil war spillovers, regional conflicts, etc.) alongside active monetary policy regimes. We assemble a comprehensive country-specific dataset of macroeconomic and financial variables that capture the key channels of policy transmission and the economic outcomes of interest.

a) Monthly sample window by country

All data are organized at a monthly frequency and aligned to a common month-end timestamp. For each country, the raw country datasets are first truncated to a common "modern" window beginning in 2011 (to capture the post-2010 geopolitical and policy environment). The effective estimation sample is determined by the intersection of (i) availability of the monetary policy shock series (meeting-month forecast surprises), (ii) availability of the country-level GPR series, and (iii) availability of the macro-financial outcomes and global controls. As a result, the stacked panel used for estimation begins in 2011 and ends in December 2024, with a reduced set of observations after enforcing complete-case availability for shocks.

b) Meeting months and shock timing conventions

Monetary policy shocks are only observed in months for which both (i) a survey-based expectation and (ii) an actual policy rate outcome are available. These months correspond to scheduled policy decision months covered by Bloomberg survey data. The policy surprise is assigned to the calendar month in which the decision occurs (and thus to that month's observation in the monthly panel). Months without a usable policy-surprise observation are not treated as zero-shocks months in the baseline estimation sample; instead, the baseline stacked panel drops remaining missing values after constructing and merging all series. Accordingly, the effective sample can be interpreted as a meeting-month panel with monthly macro-financial outcomes observed at month-end.

IV.B Variable Definitions and Transformations

a) Outcome variables

Sovereign risk premia (CDS_5Y). Sovereign risk premia are measured by the 5-year USD sovereign CDS spread, $CDS_5Y_{c,t}$, expressed in basis points (bps). This series is used in its native units; impulse responses are reported in bps.

Exchange rate ($\log(FX)$). The nominal exchange rate is defined as local currency units per U.S. dollar, $FX_{c,t}$. Under this convention, an increase in $FX_{c,t}$ corresponds to a depreciation of the local currency. The exchange-rate outcome is the log level $\log(FX_{c,t})$.

Long yield ($Y10$). The yield-curve channel is measured using the 10-year government bond yield, $Y10_{c,t}$, expressed in percent (percentage points, pp).

Credit ($\log(LOANS)$). Credit conditions are proxied by loans to the private sector, $LOANS_{c,t}$. The credit outcome is the log level $\log(LOANS_{c,t})$. That way, impulse responses represent cumulative proportional changes in the loans aggregate over the horizon.

Prices ($\log(CPI)$). The price level is measured by the consumer price index, $CPI_{c,t}$, and enters the analysis in **log levels**, $\log(CPI_{c,t})$, rather than in inflation rates.

Seasonal adjustment. When a raw series is not provided in seasonally adjusted form, we seasonally adjust it prior to constructing log-level and growth-rate variables using the U.S. Census Bureau X-13ARIMA-SEATS procedure. In the baseline dataset, this adjustment is only applied to CPI and loans for Egypt.

b) Control Variables

Inflation ($dCPI$). Monthly inflation is defined as:

$$dCPI_{c,t} \;=\; 100 \times \Delta \log(CPI_{c,t}), \tag{1}$$

computed from the (seasonally adjusted where needed) CPI series. The factor of 100 expresses inflation in percent.

Loan growth ($dLOANS$). Monthly credit growth is defined as:

$$dLOANS_{c,t} \;=\; 100 \times \Delta \log(LOANS_{c,t}), \tag{2}$$

computed from the loans series (seasonally adjusted where needed). This transformation implies that a one-unit change in $dLOANS$ corresponds to approximately one percentage point of monthly loan growth.

Currency depreciation (fx_dep). Monthly exchange-rate depreciation is defined as:

$$fx_dep_{c,t} \;=\; 100 \times \Delta \log(FX_{c,t}), \tag{3}$$

so that $fx_dep_{c,t} > 0$ corresponds to a depreciation under the local-currency-per-USD definition.

Geopolitical Risk Monthly country-specific GPR indices are obtained from the Caldara–Iacoviello country GPR series Caldara and Iacoviello (2022b).

IV.C Descriptive Statistics

This subsection documents basic distributional properties of the macro-financial variables used in the analysis and provides country-level time-series context.

Figures 1–3 plot the core macro-financial outcomes used as LP targets (sovereign CDS spreads, long yields, and the log exchange rate) together with the standardized country-level geopolitical risk index (GPRC). These panels provide two forms of context: (i) the scale and time-variation of each outcome series, and (ii) the extent to which major breaks or regime changes coincide with periods of heightened geopolitical risk.

Israel exhibits comparatively low and stable sovereign risk (CDS spreads remain at relatively modest levels for most of the sample), while long yields display a pronounced decline into the late-2010s/early-2020s followed by a sharp increase beginning in 2022. The exchange rate (log FX) is comparatively stable relative to the other countries. The GPRC series shows large spikes concentrated in the most recent period of the sample, consistent with heightened geopolitical stress episodes that are visually distinct from earlier fluctuations. Figure 1.

Turkey exhibits materially higher macro-financial volatility: CDS spreads and long yields both display wide swings, and the exchange rate shows large cumulative movements consistent with repeated depreciation episodes. Relative to Israel, the FX series is dominated by sustained trend changes rather than small cyclical fluctuations. These features motivate a state-dependent framework in which risk and credibility conditions may plausibly alter the exchange-rate and sovereign-spread channels of monetary transmission. Figure 2.

Egypt displays elevated sovereign risk relative to Israel and large exchange-rate adjustments, consistent with a macroeconomic environment in which external balance and currency dynamics can dominate short-run financial conditions. The distributional dispersion reported in Table 1 (notably the upper tail of depreciation and inflation-related measures) aligns with these visible regime shifts.

The time series suggest that large movements in sovereign risk and the exchange rate can coincide with periods of heightened geopolitical stress, motivating a state-dependent specification that allows monetary-policy transmission to vary with the geopolitical environment rather than imposing constant effects over the full sample.

V. Empirical Strategy

V.A Shock Identification

a) Identification of Monetary Policy Shocks

The first task is to construct a measure of exogenous monetary policy shocks for each country. We follow a high-frequency identification approach based on market expectations, adapted to emerging markets by Deb et al. (2023). The objective is to isolate unexpected changes in the policy rate—surprises not driven by the central bank's systematic response to the economy—so that they can be treated as exogenous shocks in our transmission analysis.

For each scheduled policy meeting in country i, we observe the announced change in the policy rate (or target rate) and compare it to the market expectation immediately prior to

the decision. Bloomberg consensus surveys of analysts' forecasts are our primary source for expectations. The monetary-policy surprise at meeting t is defined as:

$$MP\ Surprise_t = Actual\ Policy\ Rate\ Change_t - Market\ Forecast\ of\ Rate\ Change_t.$$

This surprise is the component of the rate move that was not anticipated by agents and is therefore a candidate exogenous innovation.

We map meeting-level surprises to a monthly series by assigning each surprise to the month in which the meeting occurs. If multiple meetings fall within a month (rare in our sample), we aggregate surprises within-month (by summation or averaging); in practice, nearly all months contain at most one scheduled decision.

A key refinement is to purge the raw surprises of any predictable component related to contemporaneous macroeconomic news. Even with analyst forecasts, surprises may partly reflect an endogenous reaction to publicly available information (e.g., an inflation release) that forecasters did not fully incorporate. To address this, we implement an *orthogonalization step* in the spirit of Romer and Romer (2004) and Deb et al. (2023). Specifically, we regress the forecast error on measures of macro news available just before the meeting:

$$MP\ Surprise_t = \alpha + b' Z_t + u_t,$$

where Z_t includes indicators such as inflation surprises (actual inflation relative to consensus) and real-activity surprises (e.g., GDP or monthly activity releases relative to expectations). We then take the **residual** u_t as our monetary policy shock. By construction, u_t is orthogonal to observed macro news, strengthening its interpretation as an exogenous innovation for impulse-response estimation.

This procedure closely mirrors Deb et al. (2023), who construct monetary shock series across countries by subtracting market forecasts and regressing out systematic components. It is also conceptually related to the narrative approach of Romer and Romer (2004), but relies on market-based expectations rather than central-bank internal forecasts.

We implement the procedure separately for each country, reflecting differences in data-release calendars and monetary-policy frameworks. The resulting monthly series, denoted MP_Shock_{it}, is our primary measure of an exogenous monetary tightening (+) or easing (−) shock in month t. Positive values indicate an unanticipated tightening. We use this series as an external instrument for policy changes in our regressions under standard exogeneity conditions.

Finally, we emphasize that this high-frequency approach identifies *innovations* in policy stance rather than systematic policy responses. This distinction is essential for causal inference: we seek interest-rate movements that are not simply the central bank reacting to the economy.

b) Identification of Geopolitical Risk Shocks

In parallel, we construct a series of geopolitical risk shocks for each country. We define a GPR shock as an abrupt increase in geopolitical tensions that is not driven by domestic macroeconomic conditions. The raw input is the monthly GPR index for each country, GPR_{it} (as discussed in the data section). While changes in GPR_{it} already provide a natural

proxy for geopolitical shocks—a jump in the index reflects an outbreak or escalation of conflict—we further aim to ensure exogeneity with respect to domestic economic conditions by *orthogonalizing GPR to lagged macro variables*. The motivation is that domestic economic stress can precipitate or coincide with internal political unrest that may register in the GPR index (e.g., protests or regime instability linked to economic grievances). Our objective is to filter out this potentially endogenous component and isolate shocks that are plausibly exogenous (e.g., wars, terrorist attacks, or regional conflicts not primarily driven by the domestic business cycle).

Operationally, we estimate a country-specific regression of GPR changes on lagged domestic conditions and take the residual as the **GPR shock**. For each country i:

$$\Delta GPR_{i,t} = c_i + \phi_i' X_{i,t-1} + \varepsilon_{i,t},$$

where $\Delta GPR_{i,t} = GPR_{i,t} - GPR_{i,t-1}$ is the monthly change in the geopolitical risk index and $X_{i,t-1}$ is a vector of domestic predictors lagged one month to ensure they were observed prior to the current month's risk realization. In practice, $X_{i,t-1}$ includes measures of real activity (e.g., GDP or industrial production growth), inflation, and financial conditions (e.g., credit growth), which may be correlated with domestic instability. Where relevant, we also include lagged global risk controls so that broad global risk episodes that raise GPR across countries are not mechanically attributed to country-specific shocks. The residual $\varepsilon_{i,t}$ from this regression is then our **geopolitical risk shock** series, denoted $GPR_Shock_{i,t}$.

The residual isolates the component of monthly GPR movements that is not predictable from prior domestic conditions—that is, an unanticipated jump in geopolitical risk plausibly attributable to discrete events (e.g., an outbreak of conflict, a terrorist incident, or a sudden diplomatic crisis). This identification is in the spirit of approaches that construct uncertainty shocks by purging predictable variation (Ludvigson et al., 2021). More broadly, the GPR indices themselves are designed to be externally anchored measures of geopolitical tensions based on international news coverage (Caldara and Iacoviello, 2022b), and the global index is only weakly correlated with standard macroeconomic indicators (Caldara and Iacoviello, 2022b). Consistent with this, our regressions find that lagged domestic activity and inflation explain little of the variation in GPR for these countries. Nonetheless, we remove any remaining systematic component as a conservative step.

Thus, $GPR_Shock_{i,t}$ is a continuous shock measure: large positive realizations correspond to sharp increases in geopolitical risk, while values near zero reflect normal or predictable variation. Negative realizations are also possible and correspond to easing tensions or the resolution of crises.

For comparability across countries and outcomes, we standardize $GPR_Shock_{i,t}$ (dividing by its sample standard deviation) in some specifications, allowing us to interpret effects in units of a one-standard-deviation geopolitical risk shock. These shocks can then be analyzed alongside monetary policy shocks to study whether geopolitical conditions alter the propagation of monetary-policy innovations and the broader macro-financial transmission mechanism.

c) Resulting Shock Distributions

As a basic diagnostic of the identified shock series, Figures 4–6 report the empirical distributions of the orthogonalized monetary policy shocks (MP_Shock) and geopolitical risk shocks (GPR_Shock) for each country. In each panel, the vertical reference line marks zero. Two features are salient.

First, across all three countries, the distribution of MP_Shock is sharply concentrated near zero, with comparatively few large realizations. This pattern is consistent with the identification strategy: most policy decisions are close to market expectations, while the orthogonalized residual series isolates a smaller set of genuinely unexpected policy innovations. At the same time, the MP_Shock distributions exhibit non-negligible tails on both sides, indicating occasional large surprise tightenings and easings. The tails are most pronounced for Turkey, while Israel and Egypt also display episodic outliers relative to their central mass.

Second, the GPR_Shock distributions are notably right-skewed in each country: most observations lie near zero (and frequently slightly below), while a small number of months display large positive realizations. This asymmetry is consistent with geopolitical risk being characterized by infrequent, abrupt escalations that generate large positive spikes, contrasted with more gradual normalization in calmer periods. The right tail is particularly long in Israel and Turkey, reflecting rare but sizable spikes in the measured geopolitical-risk shock.

V.B State-Dependent Local Projections

Our core model uses state-dependent local projections to estimate how the impact of a monetary policy shock on various outcome variables depends on the geopolitical risk shock. The local projection (LP) method, introduced by Jordà (2005) and widely adopted, most notably for our context, by Deb et al. (2023) is a flexible way to compute impulse response functions (IRFs) by directly regressing future values of an outcome on the shock of interest (and controls), rather than imposing a multivariate dynamic system as in VARs. We extend this to allow for an interaction between the monetary shock and the GPR shock, effectively making the response "state-dependent" on the contemporaneous risk environment.

We estimate the LP models primarily on a country-by-country basis to allow each country's dynamics to be idiosyncratic. However, we also explore a pooled panel estimation with country fixed effects for certain variables, as an additional check. In the panel version, we include country dummies and interact the shock terms with country dummies if we suspect heterogeneity in β_h across countries. In effect, this yields a panel system of equations that constrains some coefficients to be equal while allowing others (like the baseline β_h) to vary by country. Given the small number of countries (3), full pooling is not very restrictive, but we mostly rely on individual-country results to discuss heterogeneity. For inference, we use heteroskedasticity and autocorrelation robust (HAC) standard errors at the country level, and in the panel case we cluster by country to allow error dependence within each country's time series.

cumulative-change convention: For each outcome series $y_{i,t}$ (country i, month t) and horizon $h \in \{0, 1, \ldots, 12\}$, the LP left-hand side is defined as the cumulative change from

$t-1$ to $t+h$:

$$Y_{i,t}^{(h)} \;=\; y_{i,t+h} - y_{i,t-1}. \tag{4}$$

This convention is used throughout: at each horizon h, the regression coefficient on the monetary policy shock can be interpreted as the effect of a shock realized at time t on the cumulative evolution of the outcome up to $t+h$.

a) Country-level specification

For each country estimated separately, the horizon-h regression is given by:

$$\begin{aligned} Y_t^{(h)} = {} & \alpha_h + \beta_h\, MP_Shock_t + \gamma_h\, GPR_Shock_t + \delta_h \left(MP_Shock_t \times GPR_Shock_t\right) \\ & + \sum_{k=1}^{p} \phi_{h,k}\, y_{t-k} + \sum_{k=1}^{p} \left(\psi_{h,k}^{M}\, MP_Shock_{t-k} + \psi_{h,k}^{G}\, GPR_Shock_{t-k}\right) + \sum_{k=1}^{p} \kappa_{h,k}'\, C_{t-k} + \varepsilon_t^{(h)}, \end{aligned} \tag{5}$$

where $p = 4$ in the baseline country-level implementation. The regression covariances consists of:

- Time Fixed Effects α_h, implemented via monthly indicator variables
- Contemporaneous shocks and interaction: MP_Shock_t, GPR_Shock_t, and $MP_Shock_t \times GPR_Shock_t$.
- Outcome lags (levels): $y_{t-1}, \ldots, y_{t-p}$ (included in the baseline).
- Shock lags: $MP_Shock_{t-1:t-p}$ and $GPR_Shock_{t-1:t-p}$ (included in the baseline). Lagged interaction terms are not included in the baseline specification.
- Domestic controls (lags only): the baseline country routine uses the restricted set

  $$C_t \in \{dCPI_t,\ dLOANS_t,\ fx_dep_t\},$$

  and includes only $C_{t-1:t-p}$. In addition, the empirical implementation enforces that the outcome variable itself cannot enter the control set.

In the country-level procedure, the dependent variables $Y_t^{(h)}$ are constructed for all horizons first and then stacked with the regressor matrix. The estimation sample is formed via listwise deletion applied once to this stacked dataset, yielding common sample across horizons within each country, subject to a minimum-observation requirement.

Each horizon regression is estimated by OLS with heteroskedasticity- and autocorrelation-consistent (HAC/Newey–West) standard errors. The bandwidth increases with both the horizon and the lag order:

$$\texttt{maxlags} \;=\; \max\{1,\ h+p\}. \tag{6}$$

b) Pooled panel specification

In addition to country-by-country estimation, we consider a pooled panel LP with fixed effects. For each horizon h, the pooled regression is:

$$Y_{i,t}^{(h)} = \alpha_i \; + \; \tau_t + \beta_h \, MP_Shock_{i,t} + \gamma_h \, GPR_Shock_{i,t} + \delta_h \left(MP_Shock_{i,t} \times GPR_Shock_{i,t} \right) + \sum_{k=1}^{p} \phi_{h,k} \, y_{i,t-k} + \sum_{k=1}^{p} \left(\psi_{h,k}^{M} \, MP_Shock_{i,t-k} + \psi_{h,k}^{G} \, GPR_Shock_{i,t-k} \right) + \sum_{k=1}^{p} \kappa'_{h,k} \, C_{i,t-k} + \varepsilon_{i,t}^{(h)}, \tag{7}$$

where $p = 2$ in the baseline pooled implementation. The pooled model includes:

- Country fixed effects α_i, implemented via country indicator variables.
- Time fixed effects τ_t implemented via month indicator variables.
- Lag blocks mirroring the country specification: lags of the outcome in levels, lags of each shock, and lags of domestic controls.

c)Conditional Impulse Response Estimation

State dependence is governed by the interaction coefficient δ_h. For a given value g of the geopolitical-risk shock, the marginal effect of a monetary policy shock on $Y^{(h)}$ is:

$$IRF_h(g) \; = \; \beta_h + \delta_h \, g. \tag{8}$$

Following the empirical implementation, the reported conditional impulse responses evaluate g at ± 1 standard deviation (interpreting the shock series as standardized for this purpose):

$$IRF_h^{low} = \beta_h - \delta_h, \tag{9}$$

$$IRF_h^{high} = \beta_h + \delta_h. \tag{10}$$

Accordingly, the high–low difference equals:

$$IRF_h^{high} - IRF_h^{low} \; = \; 2\delta_h. \tag{11}$$

Confidence intervals for IRF_h^{low} and IRF_h^{high} are constructed using the estimated variance of $\hat{\beta}_h$, the variance of $\hat{\delta}_h$, and their covariance:

$$\mathrm{Var}(IRF_h^{low}) = \mathrm{Var}(\hat{\beta}_h) + \mathrm{Var}(\hat{\delta}_h) - 2\,\mathrm{Cov}(\hat{\beta}_h, \hat{\delta}_h), \tag{12}$$

$$\mathrm{Var}(IRF_h^{high}) = \mathrm{Var}(\hat{\beta}_h) + \mathrm{Var}(\hat{\delta}_h) + 2\,\mathrm{Cov}(\hat{\beta}_h, \hat{\delta}_h), \tag{13}$$

with critical value z set centrally in the post-processing routine used for reporting and visualization.

Testing for state dependence. In the baseline reporting, statistical evidence of state dependence at horizon h is assessed by the regression p-value on the interaction term δ_h at level $\alpha = 0.05$. A horizon is classified as exhibiting a statistically significant high–low difference when:

$$p\text{-value}(\delta_h) < \alpha,$$

which corresponds to testing the null hypothesis $H_0 : \delta_h = 0$, equivalently $H_0 : IRF_h^{high} = IRF_h^{low}$.

VI. Results

Across the three countries, statistically significant state dependence is concentrated in financial variables that map directly to the channels emphasized in the hypotheses: sovereign risk premia (CDS_5Y), long yields ($Y10$), and the exchange rate ($\log(FX)$), with additional evidence in credit ($\log(LOANS)$) and prices ($\log(CPI)$). Two broad patterns emerge.

First, several outcomes exhibit *sign reversals* across states: the conditional response to a tightening is negative under low-GPR shocks but positive under high-GPR shocks (or vice versa). This is most clearly visible in Turkey's exchange-rate and inflation responses (Tables 18 and 21) and in Israel's yield-curve responses (Table 14). Such sign reversals indicate that geopolitical conditions are not merely amplifying or attenuating average transmission, but can shift which channel dominates in the short to medium run.

Second, the horizons at which state dependence is detected differ across countries. Egypt's significant differences tend to occur at medium horizons (roughly $h = 4$ to $h = 12$) and are especially pronounced in sovereign risk and prices (Tables 7 and 11). Israel exhibits significant differences at shorter horizons for the exchange rate and credit (Tables 13 and 15) and at medium horizons for yields and CDS (Tables 14 and 12). Turkey's state dependence is concentrated at short horizons for the exchange rate and yields (Tables 18 and 19) and at medium horizons for prices (Table 21). These timing patterns are consistent with country heterogeneity in exchange-rate regimes, credibility, and the speed of pass-through from financial conditions into prices and credit.

VI.B Egypt

Sovereign risk premia: risk-premium amplification under high GPR. Egypt exhibits strong state dependence in sovereign risk premia at horizons $h = 8$ to $h = 10$ (Table 7). In low-GPR shock states, a tightening is associated with a decline in CDS spreads (e.g., $IRF_8^{low} = -86.7$ bp), while in high-GPR shock states the same tightening is associated with a substantial widening (e.g., $IRF_8^{high} = 189.5$ bp). The implied high–low contrast is large and positive at these horizons (e.g., 276 bp at $h = 8$; $p = 0.0215$), consistent with **risk-premium amplification** (H7): in months characterized by elevated geopolitical shocks, monetary tightening coincides with a materially larger increase in perceived sovereign risk.

Long yields: sign reversal and altered yield-curve transmission. For Egypt's 10-year yield, significant state dependence appears at horizons $h = 2$ to $h = 4$ and again at $h = 10$ (Table 9). The high–low differences are negative and economically meaningful (roughly -91 to

-109 bp at $h = 2$ to $h = 4$, and -168 bp at $h = 10$), reflecting a sign reversal: under low-GPR shocks, tightening increases $Y10$ (e.g., $IRF_3^{low} = 79.8$ bp), whereas under high-GPR shocks, tightening decreases $Y10$ (e.g., $IRF_3^{high} = -13.8$ bp). This pattern is not the canonical interest-rate channel prediction, and it suggests that in high-risk episodes long yields may reflect forces other than mechanical pass-through of policy tightenings (for example, changes in expected future short rates, crisis-related shifts in domestic demand for long bonds, or policy actions that coincide with stabilization efforts). At minimum, the evidence indicates that the yield-curve channel is *not invariant* to geopolitical conditions in Egypt.

Credit: muted or reversed contraction of loans under high GPR. Egypt's credit aggregate also displays significant state dependence (Table 10). At horizons $h = 4$ to $h = 7$ and $h = 12$, the high-GPR conditional response is positive while the low-GPR response is negative (e.g., at $h = 5$, $IRF^{low} = -0.046$ versus $IRF^{high} = 0.080$; high–low $= 0.125$; $p = 0.0031$). Interpreting log changes, the high–low contrast corresponds to differences on the order of 7 to 16 log points (roughly 7% to 16%) over these horizons. This pattern runs counter to the baseline credit-channel expectation of a stronger contraction in lending after tightening, and it suggests that (conditional on the identified shocks and controls) tightening shocks delivered during high-GPR episodes coincide with relatively stronger loan growth outcomes than tightening shocks delivered during low-GPR episodes. One plausible interpretation is that high-GPR months may coincide with policy packages or credibility-restoring episodes in which rate hikes accompany broader stabilization measures, improving intermediation outcomes relative to low-risk periods. Another is that measurement and sample limitations (Egypt has the smallest meeting-month sample) may make the interaction estimates sensitive at medium horizons. Regardless of mechanism, the key empirical implication is that the *credit response is state-dependent* and does not follow a uniform contractionary pattern across geopolitical environments.

Prices: substantially more inflationary response under high GPR. State dependence is strongest and most persistent for Egypt's price level (Table **??**). From $h = 2$ through $h = 12$, the high–low differences are positive and statistically significant, and they grow in magnitude at longer horizons (e.g., high–low $= 0.037$ at $h = 2$, rising to 0.145 at $h = 12$; all $p \leq 0.03$ and many $p < 0.001$). Moreover, both conditional IRFs are positive, implying that tightenings are followed by higher cumulative $\log(CPI)$ even in low-GPR states, with a markedly larger increase in high-GPR states. This finding is not consistent with a standard disinflationary monetary transmission narrative over the horizons considered and suggests that, in Egypt, geopolitical conditions can coincide with dynamics (exchange-rate pass-through, cost-push forces, or policy endogeneity that remains despite orthogonalization) that dominate the usual demand-compression mechanism within a one-year window. The most robust inference from the tables is therefore not the sign of average inflation effects, but the strong evidence that *geopolitical risk materially shifts the inflation response to monetary policy shocks.*

VI.C Israel

Exchange rate: tightening appreciates the currency under high GPR, but depreciates under low GPR. Israel exhibits clear state dependence in the exchange-rate response at short horizons $h = 0$ to $h = 4$ (Table 13). The high–low differences are negative and statistically significant, reflecting a sign reversal: in low-GPR shock states, a tightening is followed by a small depreciation (e.g., $IRF_3^{low} = 0.0084 \approx 0.84\%$), while in high-GPR shock states it is followed by an appreciation (e.g., $IRF_3^{high} = -0.0186 \approx 1.86\%$). Economically, this implies that the exchange-rate channel in Israel appears *stronger in the conventional direction during high-GPR episodes* than during low-GPR episodes, which contrasts with the attenuation prediction in H4. One interpretation is that when geopolitical shocks are salient, policy tightenings may be more strongly interpreted as a stabilizing signal (or interact with capital-flow dynamics) in a way that supports the currency in the short run.

Long yields: tightening raises yields under high GPR but lowers yields under low GPR. Israel's long yield exhibits persistent state dependence across horizons $h = 1$ to $h = 10$ (Table 14). The high–low differences are positive throughout and often large relative to Israel's yield scale (e.g., high–low = 50.8 bp at $h = 7$, $p = 0.0001$). The conditional IRFs again reverse sign: under low GPR, tightening is associated with falling long yields (e.g., $IRF_7^{low} = -29.7$ bp), whereas under high GPR it is associated with rising long yields (e.g., $IRF_7^{high} = 21.1$ bp). This pattern suggests that the yield-curve channel in Israel depends strongly on the geopolitical environment, potentially reflecting differences in term-premium behavior and/or the informational content of policy surprises across states.

Sovereign risk: modest but statistically clear risk-premium amplification. Israel's CDS results show significant high–low differences at $h = 3$ and $h = 6$ to $h = 9$ (Table 12). The magnitude is smaller than in Egypt, consistent with Israel's lower average sovereign risk, but the pattern is similar: low-GPR conditional IRFs are negative (tightening reduces CDS) while high-GPR conditional IRFs are positive (tightening increases CDS). The high–low contrasts (roughly 6 to 15 bp) provide evidence that even in Israel, geopolitical conditions can tilt the sovereign-risk response in the adverse direction, in line with the qualitative content of H7.

Credit and prices: tighter financial conditions under high GPR and a sign change in inflation. Israel's credit aggregate shows short-horizon state dependence at $h = 0$ and $h = 1$ (Table 15), with negative high–low differences indicating that tightening reduces loans more (or increases them less) under high GPR. Prices also exhibit state dependence at $h = 1$ and $h = 6$ to $h = 8$ (Table 16): low-GPR conditional IRFs are negative while high-GPR conditional IRFs are positive, implying that the inflation response to tightening is disinflationary in low-risk states but becomes inflationary in high-risk states at medium horizons. While the magnitudes are small in log points, the sign reversal is economically informative: geopolitical conditions appear to meaningfully reshape the price response even in a relatively stable macroeconomic environment.

VI.D Turkey

Exchange rate: impaired or inverted UIP channel under high GPR. Turkey shows strong evidence consistent with **attenuated (and inverted) exchange-rate transmission** under high geopolitical risk. At horizons $h = 1$ to $h = 3$, the high–low differences for $\log(FX)$ are positive and significant (Table 18), and the conditional responses reverse sign: under low GPR, tightening appreciates the currency (e.g., $IRF_2^{low} = -0.025 \approx 2.5\%$ appreciation), while under high GPR, tightening is associated with a depreciation (e.g., $IRF_2^{high} = 0.010 \approx 1.0\%$ depreciation). This is precisely the pattern anticipated by H4: when geopolitical shocks are elevated, risk-premium and flight-to-safety forces can dominate the interest-differential channel, causing the currency to weaken despite tighter policy.

Long yields: high-GPR episodes generate substantially higher long-rate responses. Turkey's long yield also exhibits pronounced state dependence at $h = 1$ to $h = 4$ (Table 19). The high–low contrasts are large (roughly 149 to 178 bp), and the conditional IRFs again flip sign: tightening lowers long yields under low GPR but raises them under high GPR. Interpreted through the term-premium lens, this is consistent with the idea that geopolitical stress increases the compensation required for holding long-duration local-currency bonds, so that in high-risk states a tightening shock is associated with a large upward movement in $Y10$ rather than the stabilization one might expect under calmer conditions.

Prices: tightening becomes inflationary under high GPR. Turkey's price level shows persistent state dependence from $h = 3$ through $h = 10$ (Table 21). The high–low differences are positive and significant at every listed horizon, with conditional IRFs that are negative under low GPR (disinflationary) but positive under high GPR (inflationary). This pattern is coherent with the exchange-rate evidence: when high GPR coincides with depreciation even after tightening, exchange-rate pass-through can plausibly dominate the demand-compression channel, resulting in higher prices following a tightening shock in the high-risk state. In other words, Turkey's results suggest a regime in which geopolitical stress can shift the economy from an interest-rate-driven stabilization mechanism to an exchange-rate-driven inflation mechanism.

VI.E Pooled panel evidence

The pooled panel estimates provide an additional summary view of state dependence across the sample (Tables 4 and 6). For $Y10$, significant high–low differences appear at short horizons $h = 0$ to $h = 4$ and again at $h = 9$ to $h = 10$. The short-horizon differences are positive (high GPR implies a higher long-yield response than low GPR), consistent with the country-level evidence that high-risk states tend to be associated with upward pressure on long yields. At longer horizons, the pooled differences become negative, indicating that the relative high-versus-low response can reverse in the medium run. For $\log(CPI)$, the pooled results show positive high–low differences at horizons $h = 2$ to $h = 4$ and again at $h = 6$ to $h = 8$ and $h = 10$, consistent with the broad cross-country tendency for high-GPR states to be associated with a more inflationary response to tightening over near-term horizons.

VI.F Synthesis and relation to hypotheses

Taken together, the results establish that monetary-policy transmission in these countries is state-dependent on contemporaneous geopolitical risk shocks, but the direction and channel through which state dependence manifests differs across institutional environments.

- **H1 (attenuated exchange-rate channel under high GPR):** strongly supported in Turkey (tightening depreciates under high GPR), but not supported in Israel, where the conventional appreciation response appears stronger under high GPR at short horizons (Table 13). Egypt does not display statistically significant state dependence in $\log(FX)$ in the reported tables, consistent with the possibility that exchange-rate management dampens detectability in meeting-month LPs.
- **H2 (risk-premium amplification):** supported in Egypt and Israel, where the CDS response to tightening is materially more adverse in high-GPR states (Tables 7 and 12).
- **Credit and inflation mechanisms (H3–H4):** evidence is mixed across countries. Israel shows a more contractionary loan response under high GPR (Table 15), consistent with impaired intermediation. Egypt's loan results instead imply weaker or reversed contraction under high GPR (Table 10). For prices, all three countries show evidence that high GPR shifts the $\log(CPI)$ response in a more inflationary direction (Tables 11, 16, and 21), with especially large magnitudes in Egypt and clear sign reversals in Israel and Turkey.

Overall, the empirical evidence supports the thesis's central premise: geopolitical risk is a first-order conditioning variable for monetary transmission in the region. In Turkey, high-GPR episodes appear to impair the exchange-rate stabilization channel and generate inflationary dynamics even after tightening. In Israel, high-GPR episodes strengthen short-run exchange-rate appreciation after tightening but are also associated with more adverse sovereign-risk dynamics and higher long yields. In Egypt, high-GPR episodes coincide with large sovereign-risk amplification and substantially more inflationary outcomes, while the yield and credit responses exhibit non-standard sign patterns that suggest the presence of additional mechanisms (or sample sensitivity) beyond the canonical transmission channels.

VII. Discussion and Limitations

This thesis asks whether monetary-policy transmission is stable across geopolitical environments. The empirical strategy estimates state-dependent impulse responses that compare the dynamic effects of an identified monetary-policy innovation under relatively low versus elevated geopolitical risk (GPR). The motivating premise is that geopolitical stress does not merely shift the level of uncertainty; it can reprice domestic risk, tighten external financing constraints, and alter portfolio allocations in ways that change the mapping from a policy-rate move to sovereign spreads, yields, exchange rates, credit, and ultimately real activity and inflation.

A natural way to interpret the results is through the channels that standard frameworks emphasize, and how those channels may be distorted in high-GPR periods. In tranquil

states, a policy-rate increase is typically expected to raise borrowing costs and risk-free yields, induce some currency appreciation, slow credit creation, and reduce inflation and activity with lags. Under geopolitical stress, however, the same policy action may operate through a different composition of margins. If heightened GPR raises risk premia and compresses risk-bearing capacity, sovereign spreads and long yields may move in ways that are less tightly linked to expected paths of short rates and more closely tied to the compensation investors require for holding domestic assets. In this case, the policy shock can transmit more strongly into sovereign risk premia—or deliver less relief through the curve in easing episodes—even when the policy innovation is measured in the same units. Similarly, if geopolitical stress impairs credit intermediation, banks may tighten lending standards or face higher funding premia, weakening the pass-through from the policy rate to effective borrowing rates and loan quantities. Finally, if risk-off dynamics and external financing conditions dominate during geopolitical turbulence, exchange-rate responses may become less consistent with interest-differential logic: currency movements can be driven by flight-to-safety and shifts in nonresident positioning rather than by the marginal change in domestic policy rates.

Cross-country comparisons help discipline these interpretations. Egypt and Turkey are natural cases where risk-premium and external-stress channels can matter disproportionately because sovereign spreads and exchange-rate movements can feed rapidly into domestic prices and credit conditions, especially when inflation is elevated or external balance constraints are salient. Israel provides an informative contrast: a more institutionally mature monetary framework and deeper markets may dampen risk-premium amplification, or at minimum shift adjustment toward expectations and communications rather than abrupt repricing in sovereign risk. The comparative evidence therefore speaks to whether geopolitical stress primarily attenuates transmission (requiring larger policy adjustments for a given stabilization objective) or redirects transmission toward external and risk-premium margins (raising the near-term financial-stability and balance-of-payments costs of tightening during stress).

The results also admit a direct policy calibration interpretation. If the impulse responses of inflation or real activity to a given policy innovation are smaller in magnitude under high GPR, then multipliers estimated from average conditions will understate the policy effort required to achieve a given stabilization objective during geopolitical stress. Conversely, if the sensitivity of sovereign spreads or the exchange rate to policy is larger in high-GPR states, then policy tightening during stress may entail greater near-term costs through financial conditions and external financing, even if the objective is to contain inflation expectations. In that sense, the state dependence documented here is not a purely academic nonlinearity; it affects the practical trade-offs central banks face in conflict-exposed environments.

More broadly, the evidence contributes to two literatures that are often treated separately. The geopolitical-risk literature documents that adverse GPR shocks are associated with weaker activity and tighter financial conditions, while the monetary-policy transmission literature characterizes average effects of policy innovations on financial and real outcomes. By conditioning monetary-policy impulse responses on geopolitical conditions, this thesis links these strands and provides evidence on a specific margin of interest to policymakers: whether the strength and timing of the policy transmission mechanism are stable when geopolitical risk is elevated. This connection is particularly relevant for emerging markets, where sovereign risk and external financing conditions are first-order determinants of macroeconomic stability, and where geopolitical events are frequent enough that "stress states" are not rare exceptions

but recurring features of the policy environment.

Several limitations qualify the interpretation of the estimates and motivate extensions. First, while forecast-based or high-frequency approaches to measuring monetary-policy innovations substantially improve identification relative to policy-rate changes in levels, they do not fully rule out information effects or imperfect measurement of the central bank's reaction to unobserved contemporaneous conditions. If the shock measure captures, in part, news about the economic outlook revealed by policymakers, impulse responses can conflate the effect of the policy action with the effect of information. Relatedly, text-based measures of geopolitical risk are externally anchored and transparent, but they inevitably pool heterogeneous events. Threats and realized acts, domestic and regional episodes, and persistent versus short-lived crises may have different macro-financial implications, and the aggregate index may smooth across them. Orthogonalization to observables mitigates concerns about mechanical correlation with domestic conditions, but it cannot guarantee that residual variation is fully exogenous to forward-looking risk repricing or anticipatory policy responses.

Second, data constraints limit both precision and comparability. Real activity is proxied at monthly frequency using available indicators, which may underrepresent services or informal activity and can be noisy in emerging markets. Financial and credit measures also differ across countries in coverage, instrument composition, and market depth; even after harmonization, some cross-country differences may partly reflect measurement rather than economics. These constraints are particularly consequential for state dependence, because identifying differential responses requires sufficient observations in both low- and high-GPR regimes, and estimates can be imprecise when high-risk periods are episodic or clustered.

Third, the state-dependent local-projection framework is designed for robustness, but it imposes structure that may be restrictive. The baseline interaction treats GPR as a continuous conditioning variable affecting the impulse response linearly; if transmission changes sharply once risk passes a threshold, or if the relevant nonlinearity is tied to crises in funding markets rather than to the level of GPR per se, the linear interaction can understate regime shifts. Moreover, although local projections are less sensitive than VARs to system misspecification, they do not fully address anticipation and dynamic feedback if expectations adjust rapidly around geopolitical events and policy actions.

Finally, reduced-form state dependence does not, by itself, uniquely identify mechanisms. Risk-premium amplification, impaired bank intermediation, and exchange-rate dominance can generate similar patterns in yields, spreads, and credit aggregates. More discriminating mechanism evidence would require additional data and decompositions, such as separating long yields into expected short-rate paths and term premia, incorporating bank-level indicators of credit supply and funding conditions, and using capital-flow and nonresident holdings data to measure external financing pressures directly. Event-type decompositions of the GPR index—distinguishing threats from acts and domestic from regional/global events—would further sharpen interpretation.

These limitations suggest clear directions for future research. Extending the analysis to nonlinear regime specifications (threshold or smooth-transition models) would test whether the interaction is better characterized as a discrete regime shift. Incorporating richer financial data—term-premium measures, capital flows, and bank-level balance-sheet indicators—would help separate risk-premium from credit-supply mechanisms and clarify when exchange-rate

movements are driven by interest differentials versus external stress. Finally, expanding the country set and stratifying results by exchange-rate regime, commodity exposure, and external vulnerability would strengthen external validity and clarify which aspects of state-dependent transmission are specific to the Eastern Mediterranean context versus general features of geopolitically exposed emerging markets.

Appendix

Summary Statistics and Time Series of Key Variables

Table 1: Summary statistics by country (outcomes and domestic controls; derived variables rebuilt from raw levels).

country	variable	N	mean	std	min	p10	p50	p90	max
EGY	CDS_5Y	70.0	592.1733	344.3911	258.7598	321.2910	477.1894	1111.8720	1835.6860
EGY	Y10	70.0	17.4512	3.3693	13.8500	14.7065	16.6000	23.9540	27.3400
EGY	dCPI	70.0	2.6066	3.2896	-2.8119	0.4978	1.4968	5.9530	19.3975
EGY	dLOANS	70.0	3.5583	9.4433	-17.3936	0.9468	2.3810	9.5549	54.4300
EGY	fx_dep	70.0	3.1041	9.6709	-3.5185	-1.1139	0.0483	9.5577	57.8699
EGY	log_CPI	70.0	4.4054	0.5545	3.6660	3.7361	4.4529	5.1959	5.4645
EGY	log_FX	70.0	2.5504	0.6315	1.7841	1.8033	2.7538	3.4304	3.9287
EGY	log_LOANS	70.0	12.5868	0.8105	11.4789	11.6179	12.4081	13.6692	13.9342
ISR	CDS_5Y	137.0	89.9831	40.4831	39.1100	50.4600	75.3900	145.2092	199.2190
ISR	Y10	137.0	2.8046	1.3303	0.6150	1.0880	2.3650	4.6280	5.2950
ISR	dCPI	137.0	0.1488	0.3435	-1.0363	-0.2284	0.1151	0.5387	1.5022
ISR	dLOANS	137.0	1.1915	1.1361	-1.9820	0.0954	1.3487	2.4338	4.6288
ISR	fx_dep	136.0	-0.0228	1.9827	-3.7857	-2.4418	-0.1700	2.5738	5.5028
ISR	log_CPI	137.0	4.4933	0.0518	4.4356	4.4550	4.4739	4.5862	4.6415
ISR	log_FX	137.0	1.2859	0.0565	1.1499	1.2059	1.2903	1.3569	1.3933
ISR	log_LOANS	137.0	2.9762	0.5066	2.2257	2.3582	2.9413	3.7391	3.8561
TUR	CDS_5Y	153.0	319.4290	153.5969	117.8090	173.9312	273.3030	548.3137	846.5370
TUR	Y10	153.0	13.3205	5.7984	6.1630	8.3050	10.6800	23.5310	28.2750
TUR	dCPI	152.0	1.7611	2.0965	-1.8399	0.0807	1.2237	3.4801	12.7503
TUR	dLOANS	153.0	2.1803	1.9692	-4.6197	0.2888	2.0765	4.0803	11.4125
TUR	fx_dep	153.0	2.0594	4.9727	-8.1325	-2.4503	1.2846	6.7508	33.8547
TUR	log_CPI	153.0	5.4978	0.7721	4.6396	4.7477	5.1907	6.8450	7.3164
TUR	log_FX	153.0	1.6556	0.9864	0.4195	0.5840	1.3279	3.3078	3.5654
TUR	log_LOANS	153.0	14.4606	0.9114	13.0181	13.3203	14.3399	15.9157	16.3321

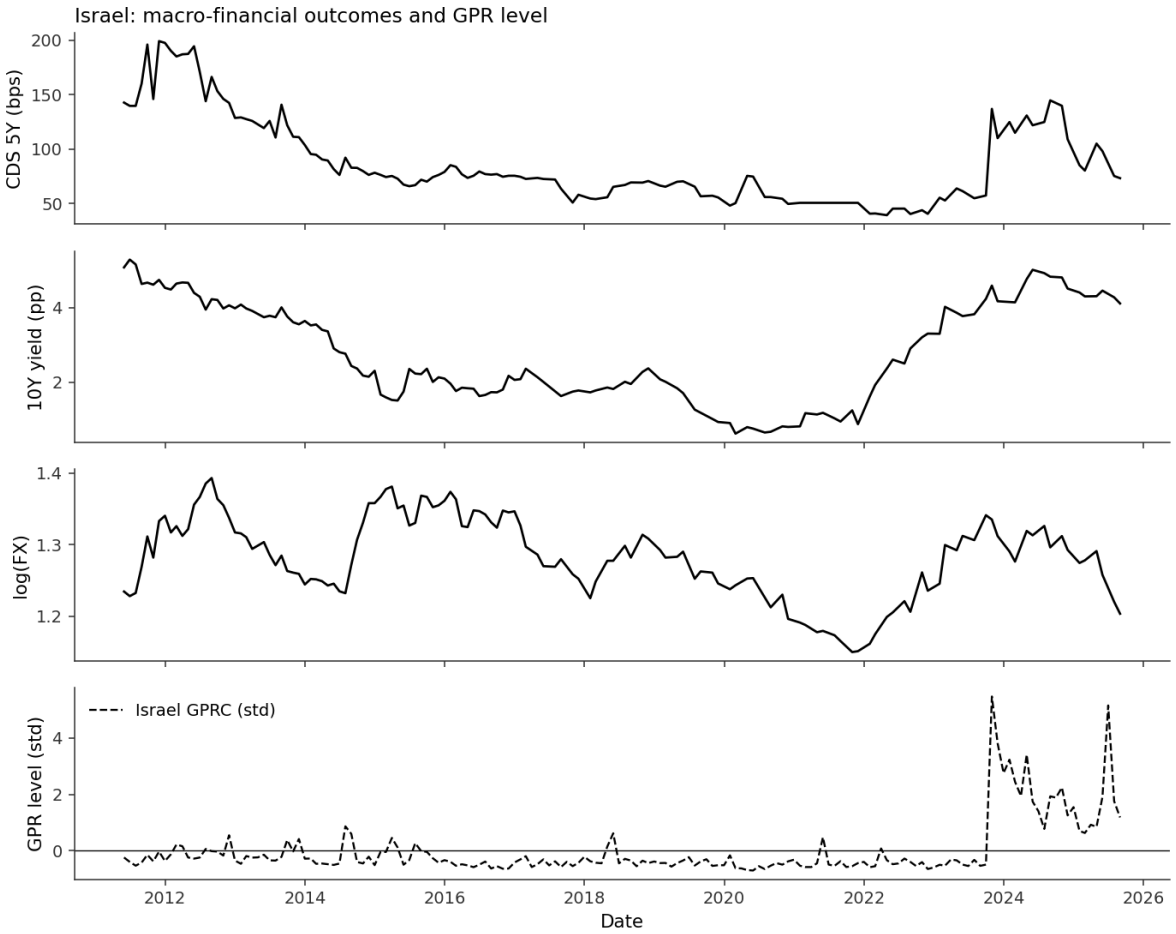

Figure 1: Israel: macro-financial outcomes and standardized geopolitical risk (GPRC). Top to bottom: CDS_5Y (bps), $Y10$ (pp), $\log(FX)$ (local currency per USD), and standardized country GPRC level.

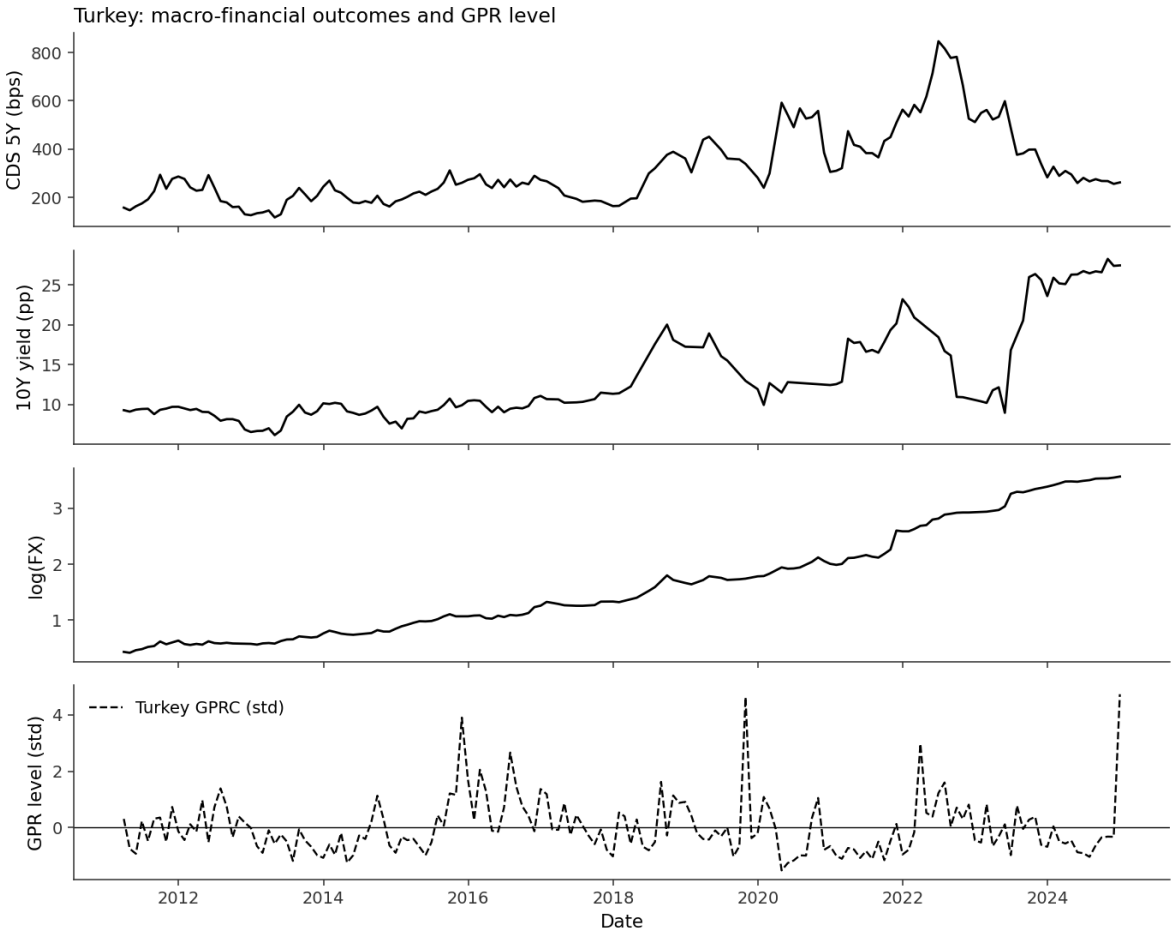

Figure 2: Turkey: macro-financial outcomes and standardized geopolitical risk (GPRC). Top to bottom: CDS_5Y (bps), $Y10$ (pp), $\log(FX)$ (local currency per USD), and standardized country GPRC level.

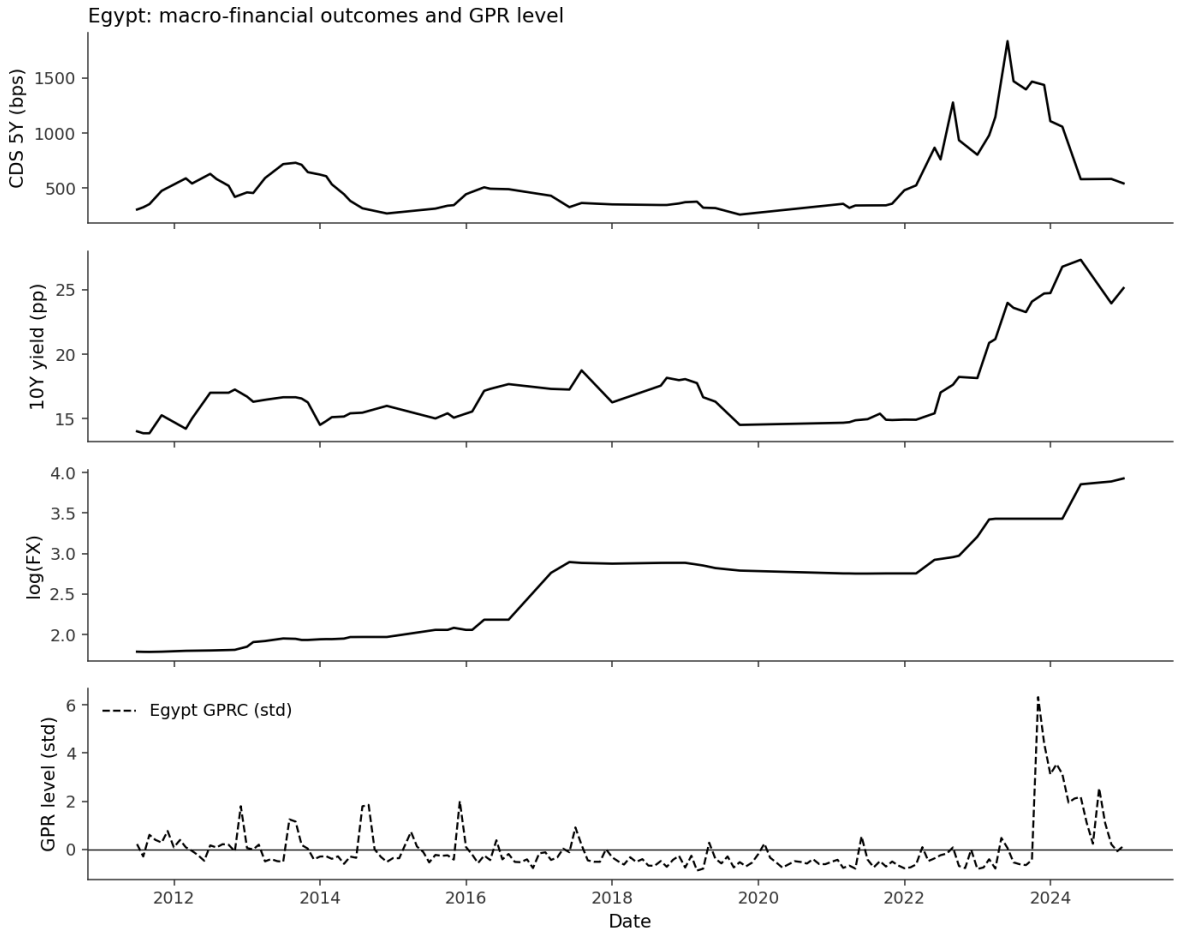

Figure 3: Egypt: macro-financial outcomes and standardized geopolitical risk (GPRC). Top to bottom: CDS_5Y (bps), $Y10$ (pp), $\log(FX)$ (local currency per USD), and standardized country GPRC level.

Constructed Shocks Distributions

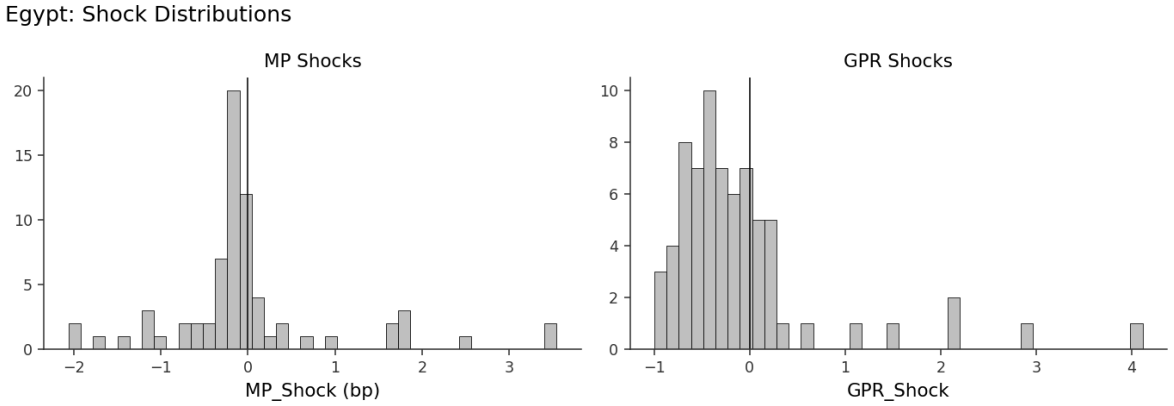

Figure 4: Egypt: empirical distributions of the identified monetary policy shock (MP_Shock) and geopolitical risk shock (GPR_Shock). The vertical line denotes zero.

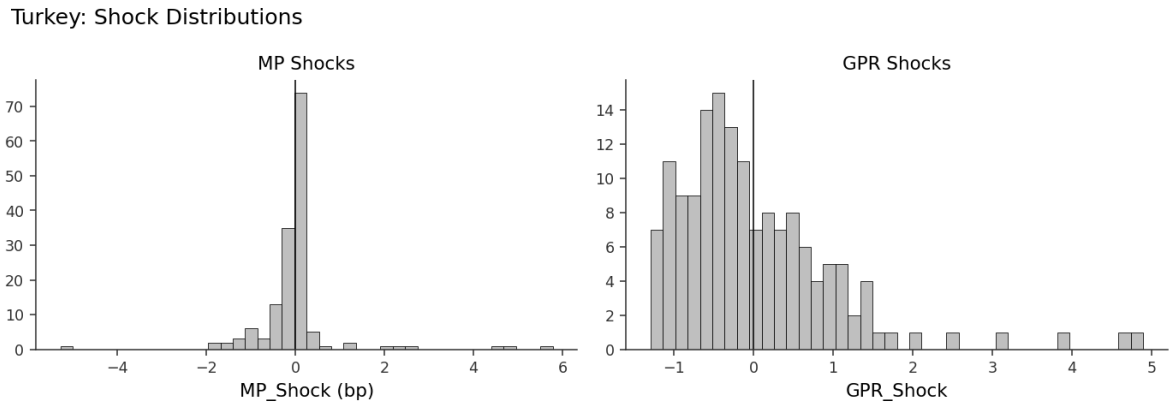

Figure 5: Turkey: empirical distributions of the identified monetary policy shock (MP_Shock) and geopolitical risk shock (GPR_Shock). The vertical line denotes zero.

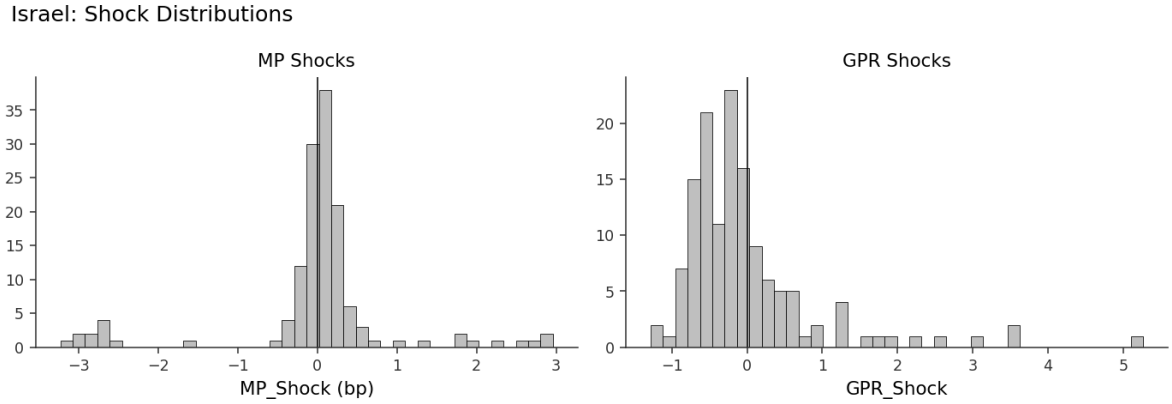

Figure 6: Israel: empirical distributions of the identified monetary policy shock (MP_Shock) and geopolitical risk shock (GPR_Shock). The vertical line denotes zero.

Pooled Panel Results

Table 2: Pooled panel: CDS_5Y (bp). Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	-17.8020	-3.3080	14.4940	[-1.7670, 30.7560]	0.1430
1	-20.2810	-16.8130	3.4680	[-22.8440, 29.7800]	0.8280
2	-17.5310	-16.8890	0.6430	[-23.5520, 24.8370]	0.9650
3	7.2340	-10.5260	-17.7600	[-47.7910, 12.2710]	0.3310
4	-14.8870	-9.6450	5.2420	[-27.6760, 38.1590]	0.7930
5	-14.9580	-6.1740	8.7840	[-21.0190, 38.5880]	0.6280
6	-29.5000	-8.5200	20.9790	[-8.7680, 50.7260]	0.2460
7	-22.1410	0.0170	22.1590	[-7.6010, 51.9180]	0.2210
8	-28.4120	-12.4380	15.9740	[-11.3600, 43.3090]	0.3360
9	-4.6710	-21.5830	-16.9120	[-42.9160, 9.0910]	0.2850
10	-16.4010	-22.6870	-6.2870	[-32.4620, 19.8890]	0.6930
11	-6.4980	-34.8240	-28.3260	[-60.5480, 3.8960]	0.1480
12	-4.5700	-29.7060	-25.1370	[-59.3090, 9.0350]	0.2260

Table 3: Pooled panel: log_FX. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	0.2400	-0.5040	-0.7450	[-2.1110, 0.6210]	0.3700
1	-0.1260	0.4040	0.5310	[-1.9340, 2.9960]	0.7230
2	-0.5820	0.5640	1.1460	[-1.4710, 3.7630]	0.4710
3	-0.9110	0.7930	1.7040	[-1.4960, 4.9040]	0.3810
4	-0.4020	0.0860	0.4880	[-2.3560, 3.3330]	0.7780
5	-0.6530	0.0050	0.6570	[-2.3230, 3.6380]	0.7170
6	0.2890	-0.6070	-0.8960	[-3.8880, 2.0960]	0.6220
7	0.2730	-0.7210	-0.9940	[-3.7950, 1.8080]	0.5600
8	0.7330	-0.2150	-0.9470	[-4.1490, 2.2550]	0.6270
9	-0.0390	-0.9650	-0.9270	[-5.3620, 3.5080]	0.7310
10	-0.1240	1.0440	1.1680	[-2.8740, 5.2100]	0.6350
11	-1.0360	0.8840	1.9200	[-2.6930, 6.5320]	0.4940
12	-0.6740	0.9320	1.6070	[-2.5030, 5.7170]	0.5200

Table 4: Pooled panel: Y10 (bp). Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	-33.8910	23.5050	57.3960	[27.6260, 87.1660]	0.0020
1	-53.2840	44.3440	97.6280	[56.6410, 138.6140]	0.0000
2	-68.7120	28.3160	97.0270	[46.9330, 147.1210]	0.0010
3	-79.5770	20.3790	99.9570	[36.9280, 162.9850]	0.0090
4	-71.8460	14.3900	86.2360	[25.2880, 147.1850]	0.0200
5	-49.1260	-8.6540	40.4720	[-16.7340, 97.6780]	0.2450
6	-54.6140	-15.8620	38.7520	[-10.0150, 87.5190]	0.1910
7	-50.1390	-26.4050	23.7330	[-31.3730, 78.8390]	0.4790
8	-41.5730	-57.0570	-15.4830	[-65.2670, 34.3010]	0.6090
9	8.2680	-79.8540	-88.1210	[-146.2180, -30.0240]	0.0130
10	16.6010	-101.6210	-118.2210	[-195.7030, -40.7400]	0.0120
11	7.7070	-118.5490	-126.2560	[-242.2450, -10.2670]	0.0730
12	22.3900	-97.5820	-119.9720	[-249.9540, 10.0110]	0.1290

Table 5: Pooled panel: log_LOANS. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	-0.3670	-0.3010	0.0660	[-0.4850, 0.6160]	0.8450
1	0.3020	-0.2780	-0.5800	[-1.3040, 0.1440]	0.1880
2	-0.3690	0.2070	0.5770	[-0.2970, 1.4510]	0.2770
3	0.1230	0.2650	0.1420	[-0.8280, 1.1110]	0.8100
4	-0.6550	-0.3790	0.2770	[-0.7820, 1.3350]	0.6670
5	-0.8470	-0.6550	0.1920	[-1.0290, 1.4140]	0.7960
6	-1.2460	-1.0820	0.1640	[-1.1190, 1.4470]	0.8340
7	-1.6720	-1.6280	0.0440	[-1.4290, 1.5180]	0.9600
8	-1.0150	-1.2630	-0.2480	[-1.6990, 1.2020]	0.7780
9	-0.1920	-1.4960	-1.3040	[-3.3240, 0.7160]	0.2880
10	-0.1720	-0.9200	-0.7480	[-2.8650, 1.3690]	0.5610
11	0.0960	-0.8420	-0.9390	[-3.6960, 1.8180]	0.5750
12	-0.6280	-0.8040	-0.1750	[-2.5880, 2.2370]	0.9050

Table 6: Pooled panel: log_CPI. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	0.1780	0.4820	0.3040	[-0.1310, 0.7400]	0.2500
1	0.1520	0.5820	0.4290	[-0.1850, 1.0430]	0.2500
2	-0.2710	1.2270	1.4990	[0.5560, 2.4420]	0.0090
3	-0.0910	1.4610	1.5520	[0.4170, 2.6870]	0.0240
4	-0.3720	1.1220	1.4940	[0.3590, 2.6290]	0.0300
5	-0.0350	1.2250	1.2600	[0.1210, 2.3990]	0.0690
6	-0.4930	1.1910	1.6840	[0.4600, 2.9070]	0.0240
7	-0.6410	1.3790	2.0190	[0.7480, 3.2900]	0.0090
8	-0.8960	1.5580	2.4540	[0.9550, 3.9530]	0.0070
9	-0.1390	1.9110	2.0490	[-0.1450, 4.2440]	0.1240
10	-0.3680	2.7200	3.0870	[0.8620, 5.3130]	0.0220
11	-0.2130	1.8990	2.1130	[-0.1960, 4.4220]	0.1320
12	-0.2600	1.6260	1.8870	[-0.5590, 4.3330]	0.2040

Egypt Results

Table 7: Egypt: CDS_5Y (bp). Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	0.0218	33.1460	33.1242	[-47.4450, 113.6933]	0.4204
1	-13.2277	-0.6752	12.5525	[-78.6053, 103.7104]	0.7872
2	79.0482	20.3626	-58.6856	[-208.3660, 90.9948]	0.4422
3	59.0225	12.1012	-46.9213	[-257.8716, 164.0290]	0.6629
4	-31.0626	28.7766	59.8392	[-200.5583, 320.2367]	0.6524
5	-99.5906	90.3215	189.9121	[-95.5226, 475.3468]	0.1922
6	-101.3636	84.4433	185.8069	[-92.6548, 464.2687]	0.1909
7	-0.9184	104.4399	105.3584	[-190.0049, 400.7217]	0.4845
8	-86.6761	189.4582	276.1343	[40.6376, 511.6310]	0.0215*
9	-128.3919	118.5685	246.9604	[11.9249, 481.9960]	0.0395*
10	-160.9744	39.5329	200.5072	[23.8829, 377.1315]	0.0261*
11	-150.9498	-67.2913	83.6584	[-73.0123, 240.3292]	0.2953
12	-140.3437	-152.0206	-11.6769	[-127.0632, 103.7093]	0.8428

Table 8: Egypt: log_FX. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	0.0600	0.0157	-0.0443	[-0.1005, 0.0119]	0.1223
1	0.0640	-0.0105	-0.0745	[-0.1656, 0.0165]	0.1086
2	0.0810	-0.0198	-0.1008	[-0.2294, 0.0277]	0.1242
3	0.1387	-0.0158	-0.1545	[-0.3221, 0.0131]	0.0708
4	0.1656	-0.0390	-0.2045	[-0.4246, 0.0155]	0.0685
5	0.1964	-0.0601	-0.2565	[-0.5211, 0.0080]	0.0574
6	0.1847	-0.0227	-0.2074	[-0.4501, 0.0353]	0.0940
7	0.1637	0.0048	-0.1588	[-0.3548, 0.0371]	0.1121
8	0.1339	0.0110	-0.1229	[-0.3338, 0.0880]	0.2533
9	0.1218	0.0273	-0.0945	[-0.2996, 0.1106]	0.3666
10	0.1349	0.1288	-0.0061	[-0.2662, 0.2540]	0.9633
11	0.1255	0.1816	0.0561	[-0.2743, 0.3864]	0.7394
12	0.0797	0.2198	0.1401	[-0.2311, 0.5112]	0.4594

Table 9: Egypt: Y10 (bp). Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	21.4206	2.2115	-19.2091	[-67.8769, 29.4588]	0.4392
1	54.2064	-3.1833	-57.3897	[-116.0555, 1.2760]	0.0552
2	55.8822	-35.1730	-91.0552	[-175.5828, -6.5276]	0.0347*
3	79.7884	-13.8187	-93.6071	[-177.1359, -10.0784]	0.0281*
4	37.8995	-71.4553	-109.3548	[-197.5837, -21.1259]	0.0151*
5	44.5281	-72.4958	-117.0239	[-244.1659, 10.1181]	0.0712
6	10.1553	-117.8983	-128.0535	[-264.6644, 8.5574]	0.0662
7	26.1536	-78.4810	-104.6346	[-236.5595, 27.2902]	0.1201
8	9.8923	-90.0454	-99.9377	[-219.5635, 19.6880]	0.1015
9	-9.0556	-129.8772	-120.8215	[-244.1800, 2.5369]	0.0549
10	-4.4195	-171.9422	-167.5227	[-313.9547, -21.0906]	0.0249*
11	-49.3978	-158.5929	-109.1952	[-241.1248, 22.7345]	0.1048
12	-54.3660	-193.4421	-139.0761	[-340.7368, 62.5847]	0.1765

Table 10: Egypt: log_LOANS. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	0.0417	-0.0033	-0.0450	[-0.1130, 0.0230]	0.1950
1	-0.0159	0.0129	0.0288	[-0.0400, 0.0976]	0.4114
2	-0.0206	0.0259	0.0465	[-0.0444, 0.1374]	0.3163
3	0.0103	0.0118	0.0015	[-0.1160, 0.1190]	0.9805
4	-0.0625	0.0127	0.0753	[0.0073, 0.1433]	0.0300*
5	-0.0456	0.0798	0.1254	[0.0423, 0.2085]	0.0031*
6	-0.0390	0.0541	0.0931	[0.0086, 0.1776]	0.0308*
7	-0.0508	0.0456	0.0964	[0.0035, 0.1894]	0.0421*
8	-0.0435	0.0426	0.0861	[-0.0280, 0.2001]	0.1390
9	-0.0362	0.0242	0.0604	[-0.0795, 0.2004]	0.3973
10	-0.0261	0.0365	0.0626	[-0.1014, 0.2265]	0.4544
11	-0.0166	0.0684	0.0850	[-0.0640, 0.2341]	0.2634
12	-0.0617	0.1008	0.1625	[0.0085, 0.3165]	0.0386*

Table 11: Egypt: log_CPI. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	0.0078	0.0180	0.0102	[-0.0087, 0.0290]	0.2916
1	0.0127	0.0241	0.0114	[-0.0143, 0.0372]	0.3847
2	0.0117	0.0485	0.0368	[0.0100, 0.0636]	0.0071*
3	0.0265	0.0641	0.0376	[0.0184, 0.0567]	0.0001*
4	0.0277	0.0773	0.0496	[0.0189, 0.0803]	0.0015*
5	0.0401	0.0865	0.0464	[-0.0010, 0.0937]	0.0551
6	0.0382	0.1002	0.0620	[0.0150, 0.1090]	0.0097*
7	0.0426	0.1192	0.0766	[0.0290, 0.1241]	0.0016*
8	0.0407	0.1130	0.0723	[0.0071, 0.1374]	0.0296*
9	0.0493	0.1126	0.0633	[-0.0024, 0.1290]	0.0589
10	0.0275	0.1483	0.1208	[0.0527, 0.1890]	0.0005*
11	0.0319	0.1545	0.1227	[0.0569, 0.1884]	0.0003*
12	0.0094	0.1543	0.1449	[0.0691, 0.2208]	0.0002*

Israel Results

Table 12: Israel: CDS_5Y (bp). Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	-0.3165	-0.1675	0.1490	[-3.0462, 3.3442]	0.9272
1	-2.9756	0.6009	3.5765	[-0.2708, 7.4238]	0.0684
2	-2.5609	2.2206	4.7815	[-0.4627, 10.0256]	0.0739
3	-3.2960	2.6113	5.9073	[2.2994, 9.5152]	0.0013*
4	-3.0373	3.0480	6.0853	[-0.1088, 12.2794]	0.0542
5	-0.9472	3.7739	4.7211	[-1.1095, 10.5516]	0.1125
6	-2.6516	3.7999	6.4516	[1.5400, 11.3631]	0.0100*
7	-10.7283	4.0882	14.8166	[5.9188, 23.7143]	0.0011*
8	-7.1346	3.0566	10.1912	[2.1940, 18.1884]	0.0125*
9	-5.5161	4.7628	10.2789	[1.4399, 19.1178]	0.0226*
10	-4.5633	2.8320	7.3954	[-0.2828, 15.0735]	0.0591
11	-2.0819	2.9906	5.0725	[-1.1180, 11.2631]	0.1083
12	-1.9908	2.8163	4.8071	[-2.6834, 12.2976]	0.2084

Table 13: Israel: log_FX. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	0.0038	-0.0034	-0.0072	[-0.0133, -0.0011]	0.0211*
1	0.0054	-0.0093	-0.0146	[-0.0248, -0.0045]	0.0048*
2	0.0068	-0.0139	-0.0207	[-0.0355, -0.0059]	0.0061*
3	0.0084	-0.0186	-0.0270	[-0.0417, -0.0124]	0.0003*
4	0.0057	-0.0157	-0.0214	[-0.0402, -0.0026]	0.0259*
5	0.0037	-0.0129	-0.0166	[-0.0352, 0.0019]	0.0788
6	-0.0025	-0.0146	-0.0121	[-0.0291, 0.0050]	0.1645
7	-0.0019	-0.0155	-0.0136	[-0.0323, 0.0051]	0.1540
8	-0.0004	-0.0132	-0.0128	[-0.0306, 0.0050]	0.1584
9	-0.0032	-0.0104	-0.0072	[-0.0255, 0.0111]	0.4398
10	-0.0013	-0.0106	-0.0093	[-0.0284, 0.0098]	0.3385
11	-0.0071	-0.0113	-0.0041	[-0.0218, 0.0135]	0.6453
12	-0.0090	-0.0191	-0.0102	[-0.0267, 0.0063]	0.2264

Table 14: Israel: Y10 (bp). Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	-4.2258	2.4251	6.6509	[-0.2491, 13.5508]	0.0589
1	-6.4790	4.8306	11.3095	[0.9300, 21.6890]	0.0327*
2	-10.5005	5.6205	16.1210	[4.7186, 27.5234]	0.0056*
3	-10.5882	9.4941	20.0823	[8.6382, 31.5264]	0.0006*
4	-9.5531	10.2448	19.7979	[4.4299, 35.1658]	0.0116*
5	-16.9561	16.3242	33.2802	[14.3037, 52.2568]	0.0006*
6	-21.1823	21.3310	42.5133	[20.5344, 64.4923]	0.0001*
7	-29.7369	21.0902	50.8272	[25.3375, 76.3168]	0.0001*
8	-25.6097	21.1147	46.7244	[20.3213, 73.1274]	0.0005*
9	-22.2389	17.7424	39.9813	[10.8107, 69.1519]	0.0072*
10	-20.7648	11.9955	32.7604	[0.6577, 64.8630]	0.0455*
11	-16.5556	7.6167	24.1723	[-12.1085, 60.4531]	0.1916
12	-16.5168	11.4322	27.9490	[-9.9487, 65.8468]	0.1483

Table 15: Israel: log_LOANS. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	0.0038	-0.0019	-0.0057	[-0.0100, -0.0014]	0.0090*
1	0.0047	-0.0032	-0.0079	[-0.0158, -0.0000]	0.0497*
2	0.0035	-0.0036	-0.0071	[-0.0183, 0.0042]	0.2175
3	-0.0006	-0.0020	-0.0014	[-0.0144, 0.0116]	0.8296
4	0.0004	-0.0001	-0.0005	[-0.0163, 0.0153]	0.9498
5	-0.0005	-0.0014	-0.0009	[-0.0204, 0.0186]	0.9272
6	0.0011	-0.0028	-0.0038	[-0.0266, 0.0190]	0.7426
7	0.0035	-0.0046	-0.0081	[-0.0345, 0.0182]	0.5450
8	0.0060	-0.0057	-0.0118	[-0.0415, 0.0180]	0.4384
9	0.0047	-0.0079	-0.0127	[-0.0471, 0.0217]	0.4708
10	0.0016	-0.0095	-0.0111	[-0.0486, 0.0264]	0.5614
11	-0.0024	-0.0112	-0.0088	[-0.0484, 0.0307]	0.6614
12	-0.0040	-0.0136	-0.0096	[-0.0511, 0.0319]	0.6517

Table 16: Israel: log_CPI. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	-0.0007	-0.0000	0.0006	[-0.0004, 0.0017]	0.2381
1	-0.0013	0.0008	0.0022	[0.0004, 0.0040]	0.0191*
2	-0.0012	0.0008	0.0021	[-0.0001, 0.0042]	0.0630
3	-0.0010	0.0009	0.0019	[-0.0009, 0.0047]	0.1757
4	-0.0010	0.0013	0.0023	[-0.0011, 0.0057]	0.1855
5	-0.0011	0.0017	0.0027	[-0.0010, 0.0065]	0.1551
6	-0.0019	0.0026	0.0044	[0.0002, 0.0087]	0.0422*
7	-0.0024	0.0034	0.0058	[0.0013, 0.0103]	0.0118*
8	-0.0022	0.0027	0.0049	[0.0004, 0.0094]	0.0340*
9	-0.0019	0.0025	0.0044	[-0.0006, 0.0094]	0.0865
10	-0.0016	0.0027	0.0043	[-0.0013, 0.0098]	0.1349
11	-0.0019	0.0030	0.0049	[-0.0016, 0.0114]	0.1380
12	-0.0014	0.0036	0.0050	[-0.0021, 0.0120]	0.1653

Turkey Results

Table 17: Turkey: CDS_5Y (bp). Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	9.5425	-2.3838	-11.9263	[-32.4282, 8.5756]	0.2542
1	4.5532	8.3911	3.8379	[-32.6529, 40.3287]	0.8367
2	-3.6991	15.7368	19.4360	[-21.5546, 60.4265]	0.3527
3	-16.9048	1.8984	18.8032	[-21.9037, 59.5101]	0.3653
4	-13.1283	-36.3968	-23.2685	[-62.4057, 15.8687]	0.2439
5	-19.7460	-40.2808	-20.5348	[-55.2167, 14.1471]	0.2458
6	-19.9569	-41.7831	-21.8262	[-69.8388, 26.1864]	0.3729
7	-23.1276	-48.7524	-25.6248	[-70.6531, 19.4035]	0.2647
8	-22.3599	-39.9042	-17.5443	[-65.3888, 30.3002]	0.4723
9	-27.4814	-46.6012	-19.1198	[-70.9864, 32.7468]	0.4700
10	-21.9692	-32.8680	-10.8988	[-65.2024, 43.4049]	0.6940
11	-21.5023	-15.3538	6.1486	[-41.8760, 54.1731]	0.8019
12	-23.6125	0.0110	23.6236	[-23.1239, 70.3711]	0.3219

Table 18: Turkey: log_FX. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	-0.0103	0.0030	0.0133	[-0.0115, 0.0382]	0.2922
1	-0.0230	0.0114	0.0345	[0.0080, 0.0610]	0.0108*
2	-0.0252	0.0101	0.0353	[0.0025, 0.0682]	0.0352*
3	-0.0292	0.0094	0.0387	[0.0025, 0.0748]	0.0360*
4	-0.0303	-0.0012	0.0291	[-0.0132, 0.0714]	0.1775
5	-0.0350	-0.0008	0.0342	[-0.0056, 0.0740]	0.0922
6	-0.0375	-0.0016	0.0359	[-0.0078, 0.0796]	0.1077
7	-0.0393	0.0013	0.0407	[-0.0069, 0.0882]	0.0938
8	-0.0339	0.0008	0.0347	[-0.0110, 0.0805]	0.1368
9	-0.0370	-0.0107	0.0263	[-0.0223, 0.0749]	0.2881
10	-0.0313	-0.0202	0.0111	[-0.0338, 0.0560]	0.6270
11	-0.0289	-0.0253	0.0037	[-0.0452, 0.0525]	0.8831
12	-0.0231	-0.0152	0.0078	[-0.0380, 0.0536]	0.7371

Table 19: Turkey: Y10 (bp). Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	-48.7562	14.8504	63.6065	[-16.7705, 143.9836]	0.1209
1	-85.8883	62.8445	148.7328	[54.0940, 243.3717]	0.0021*
2	-102.9470	62.1053	165.0524	[52.5226, 277.5822]	0.0040*
3	-158.2559	19.8864	178.1424	[28.6076, 327.6772]	0.0195*
4	-148.1507	15.9409	164.0916	[7.1500, 321.0333]	0.0404*
5	-116.0914	-50.3373	65.7540	[-106.2227, 237.7308]	0.4536
6	-94.6732	-58.6458	36.0274	[-104.4299, 176.4846]	0.6151
7	-85.3189	-58.5096	26.8093	[-148.7765, 202.3952]	0.7647
8	-71.3861	-30.1719	41.2142	[-112.6847, 195.1131]	0.5997
9	-61.6448	-58.0083	3.6364	[-169.1240, 176.3969]	0.9671
10	-62.5309	-52.5061	10.0248	[-161.9001, 181.9497]	0.9090
11	-45.4624	-73.3282	-27.8658	[-194.6171, 138.8856]	0.7433
12	-26.8974	-72.6181	-45.7207	[-224.5158, 133.0745]	0.6162

Table 20: Turkey: log_LOANS. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	-0.0012	-0.0032	-0.0020	[-0.0082, 0.0043]	0.5327
1	-0.0065	-0.0062	0.0003	[-0.0079, 0.0086]	0.9395
2	-0.0091	-0.0071	0.0020	[-0.0114, 0.0153]	0.7719
3	-0.0112	-0.0114	-0.0001	[-0.0171, 0.0168]	0.9888
4	-0.0109	-0.0225	-0.0116	[-0.0324, 0.0092]	0.2761
5	-0.0144	-0.0271	-0.0127	[-0.0346, 0.0092]	0.2547
6	-0.0164	-0.0331	-0.0167	[-0.0440, 0.0105]	0.2289
7	-0.0196	-0.0364	-0.0167	[-0.0474, 0.0140]	0.2850
8	-0.0217	-0.0377	-0.0160	[-0.0470, 0.0151]	0.3130
9	-0.0271	-0.0359	-0.0088	[-0.0440, 0.0263]	0.6226
10	-0.0266	-0.0385	-0.0119	[-0.0483, 0.0245]	0.5222
11	-0.0287	-0.0376	-0.0089	[-0.0463, 0.0286]	0.6415
12	-0.0290	-0.0346	-0.0056	[-0.0422, 0.0310]	0.7652

Table 21: Turkey: log_CPI. Full horizon results.

h	IRF Low	IRF High	High–Low	95% CI (High–Low)	p
0	0.0011	-0.0002	-0.0013	[-0.0072, 0.0047]	0.6692
1	-0.0058	0.0001	0.0059	[-0.0035, 0.0152]	0.2177
2	-0.0132	0.0016	0.0147	[-0.0000, 0.0295]	0.0503
3	-0.0177	0.0054	0.0231	[0.0061, 0.0401]	0.0077*
4	-0.0207	0.0058	0.0265	[0.0062, 0.0468]	0.0106*
5	-0.0221	0.0098	0.0319	[0.0071, 0.0567]	0.0118*
6	-0.0235	0.0112	0.0347	[0.0044, 0.0650]	0.0250*
7	-0.0271	0.0135	0.0406	[0.0079, 0.0732]	0.0148*
8	-0.0294	0.0142	0.0436	[0.0091, 0.0780]	0.0132*
9	-0.0289	0.0144	0.0433	[0.0088, 0.0778]	0.0138*
10	-0.0288	0.0106	0.0394	[0.0021, 0.0767]	0.0387*
11	-0.0294	0.0076	0.0370	[-0.0009, 0.0749]	0.0554
12	-0.0281	0.0029	0.0310	[-0.0096, 0.0717]	0.1349

Israel: Impulse Responses

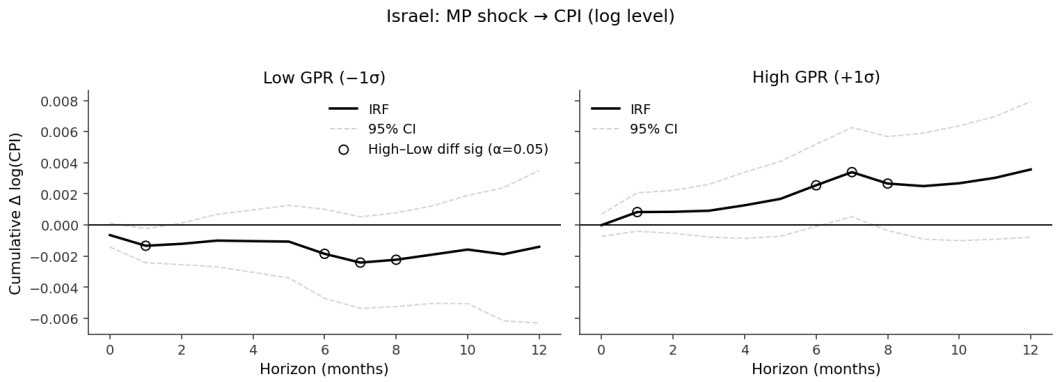

Figure 7: Israel: Impulse response of prices, $\log(CPI)$, to a monetary policy shock.

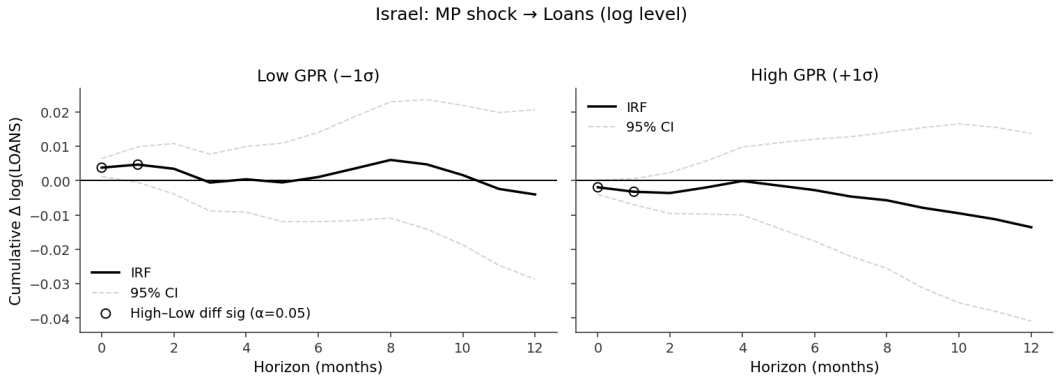

Figure 8: Israel: Impulse response of credit, $\log(LOANS)$, to a monetary policy shock.

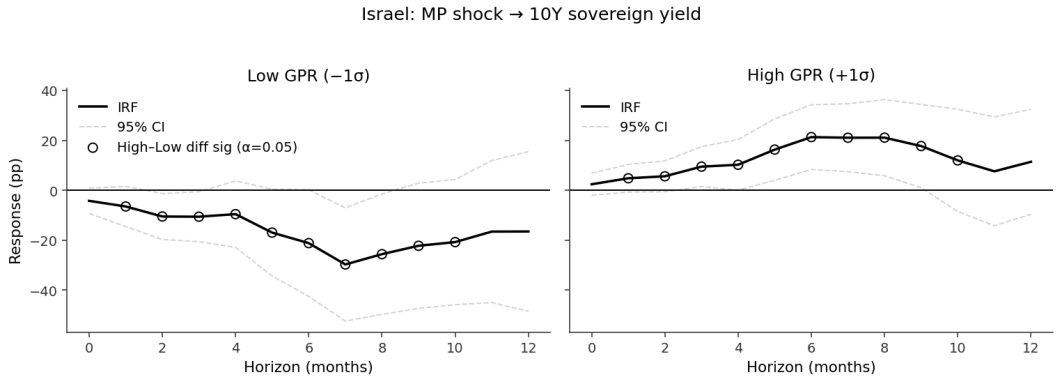

Figure 9: Israel: Impulse response of the 10-year yield, $Y10$, to a monetary policy shock.

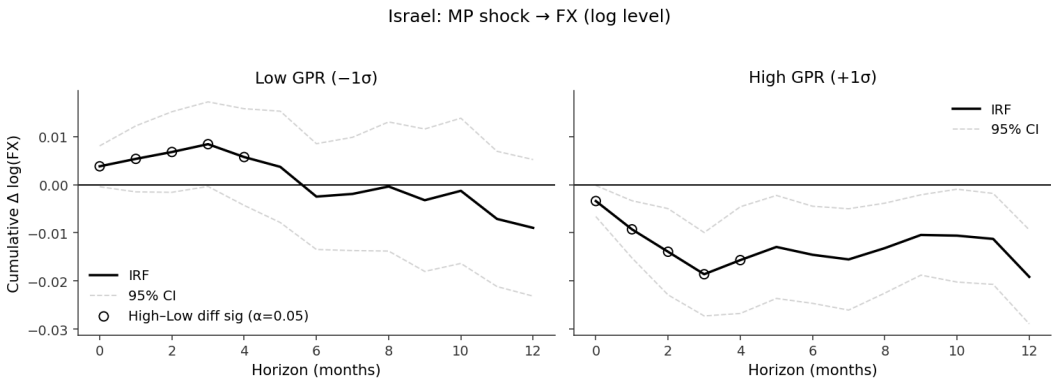

Figure 10: Israel: Impulse response of the exchange rate, $\log(FX)$, to a monetary policy shock.

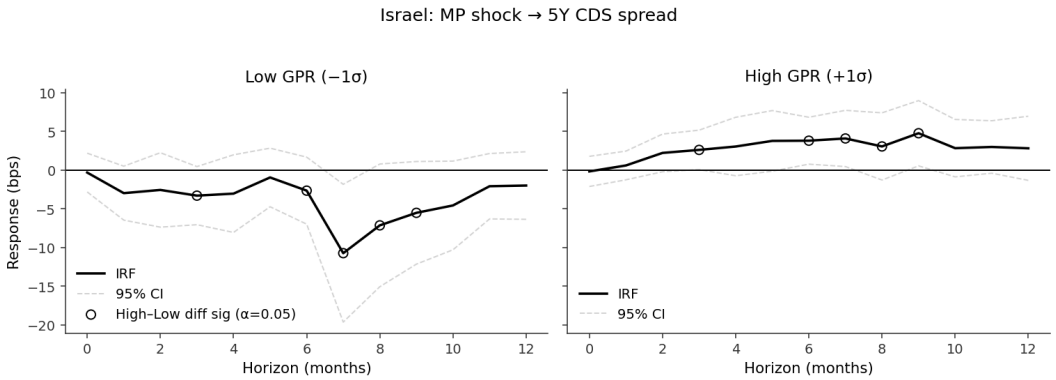

Figure 11: Israel: Impulse response of sovereign credit risk, CDS_5Y, to a monetary policy shock.

Egypt: Impulse Responses

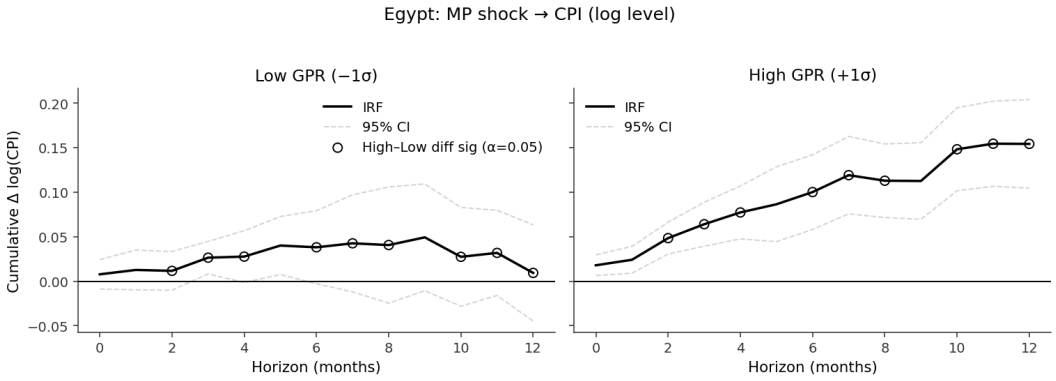

Figure 12: Egypt: Impulse response of prices, $\log(CPI)$, to a monetary policy shock.

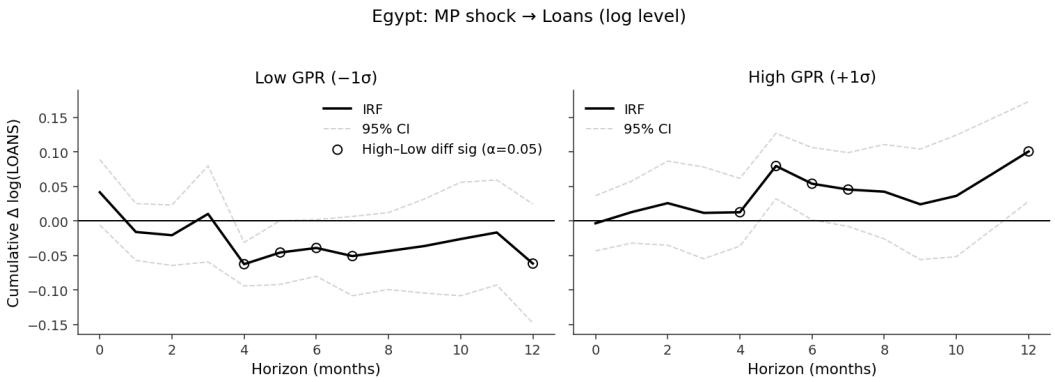

Figure 13: Egypt: Impulse response of credit, $\log(LOANS)$, to a monetary policy shock.

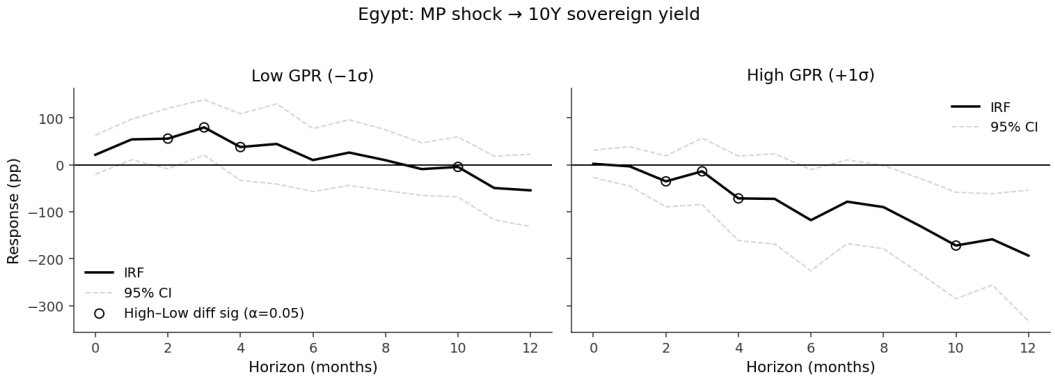

Figure 14: Egypt: Impulse response of the 10-year yield, $Y10$, to a monetary policy shock.

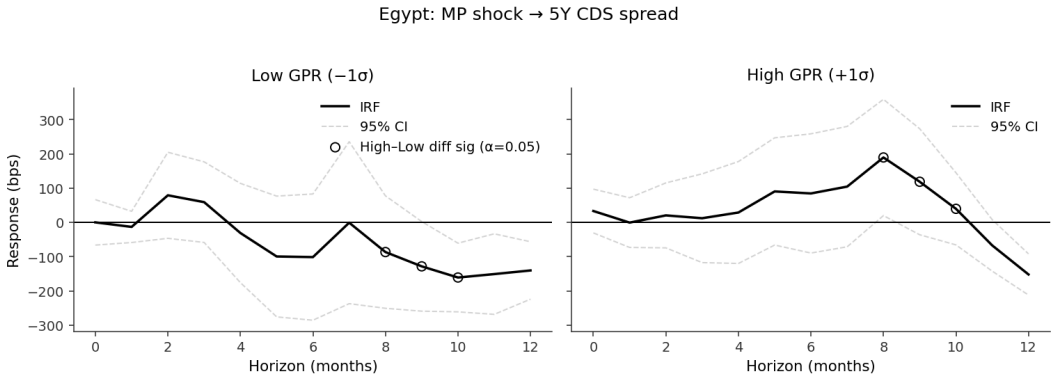

Figure 15: Egypt: Impulse response of sovereign credit risk, CDS_5Y, to a monetary policy shock.

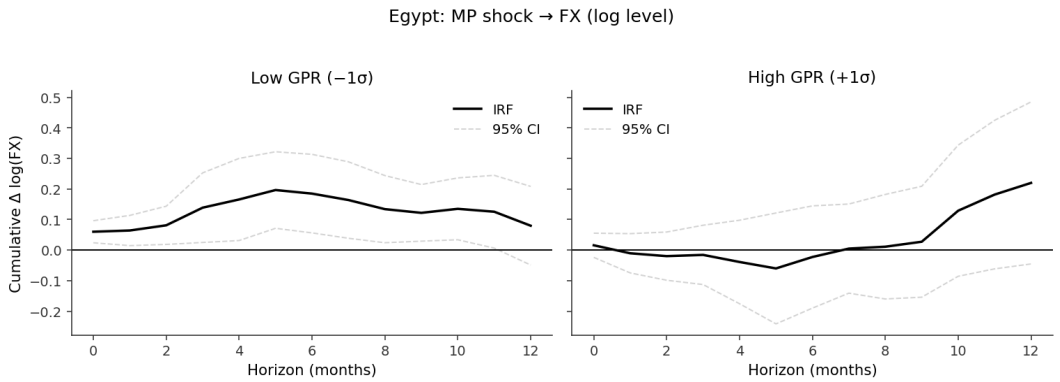

Figure 16: Egypt: Impulse response of the exchange rate, $\log(FX)$, to a monetary policy shock.

Turkey: Impulse Responses

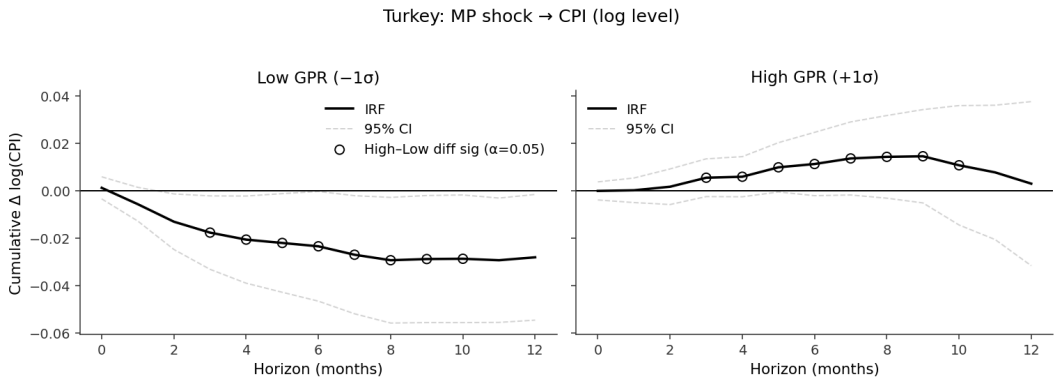

Figure 17: Turkey: Impulse response of prices, $\log(CPI)$, to a monetary policy shock.

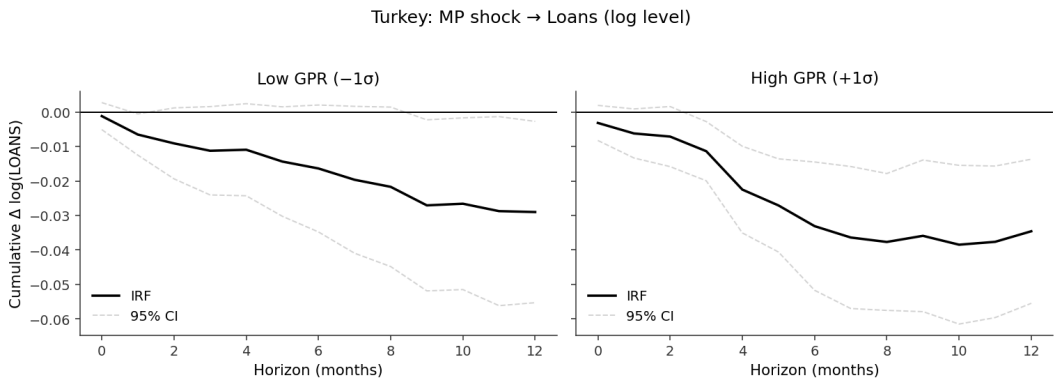

Figure 18: Turkey: Impulse response of credit, $\log(LOANS)$, to a monetary policy shock.

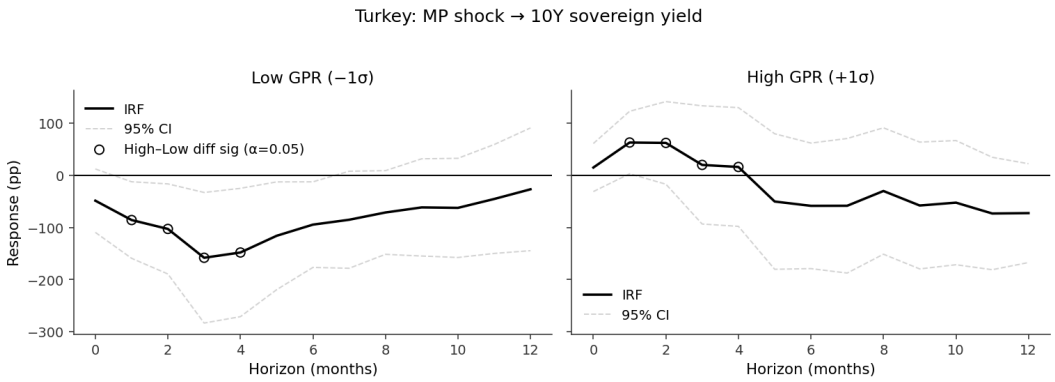

Figure 19: Turkey: Impulse response of the 10-year yield, $Y10$, to a monetary policy shock.

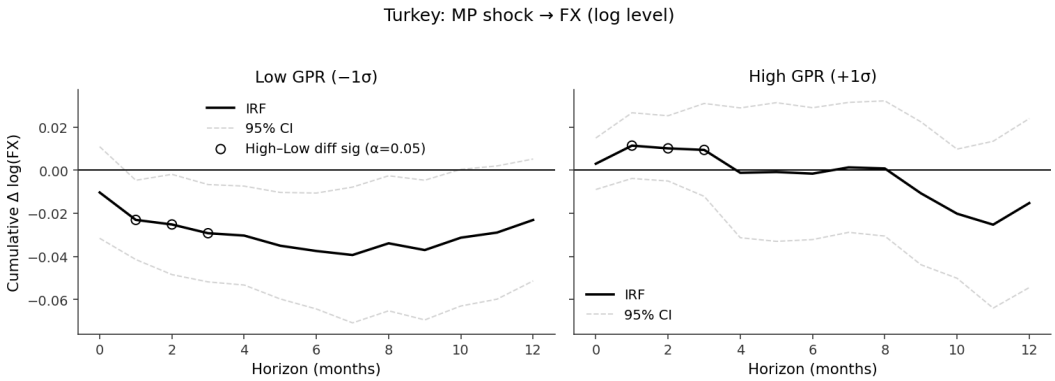

Figure 20: Turkey: Impulse response of the exchange rate, $\log(FX)$, to a monetary policy shock.

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